Chapter 5. Biopotential Electrodes

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ÇANKAYA ÜNİVERSİTESİ
MEKATRONİK MÜHENDİSLİĞİ BÖLÜMÜ

Biopotential Electrodes

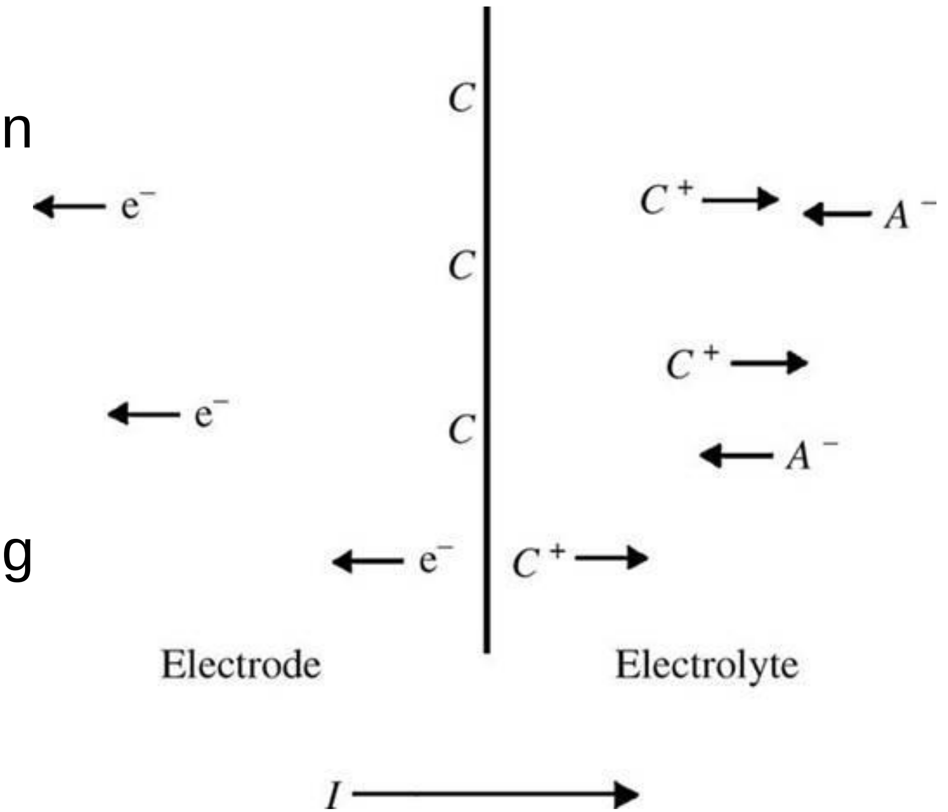
- Interface between the body and the electronic measuring apparatus to measure and record potentials.
- Biopotential electrodes conduct current across the interface between the body and the electronic measuring circuit, as other potential measurement devidces do.
- Electrode carries out a transducing function, because current is carried in the body by ions, whereas it is carried in the electrode and its lead wire by electrons.
- Electrode serve as a transducer to change an ionic current into an electronic current.
- Electrodes have electrical characteristics and can be modelled with equivalent circuits based on this characteristics.

Electrode-Electrolyte interface

- To understand the passage of electric current from the body to an electrode, electrode-electrolyte interface is examined.

The faradaic current is the current generated by the reduction or oxidation of some chemical substance at an electrode.

- A net current, I, that crosses the interface, passing from the electrode to the electrolyte, consists of
- (1) electrons moving in a direction opposite to that of the current **<u>in the electrode</u>**,
- (2) cations (denoted by C^+) moving in the same direction as the current, and
- (3) anions (denoted by A^-) moving in a direction opposite to that of the current **<u>in the electrolyte</u>**.



- For charge to cross the interface—***there are no free electrons in the electrolyte*** and ***no free cations or anions in the electrode***—

$$C \rightleftarrows C^{n+} + ne^{-}$$

where *n* is the valence of *C.*

Assumption:

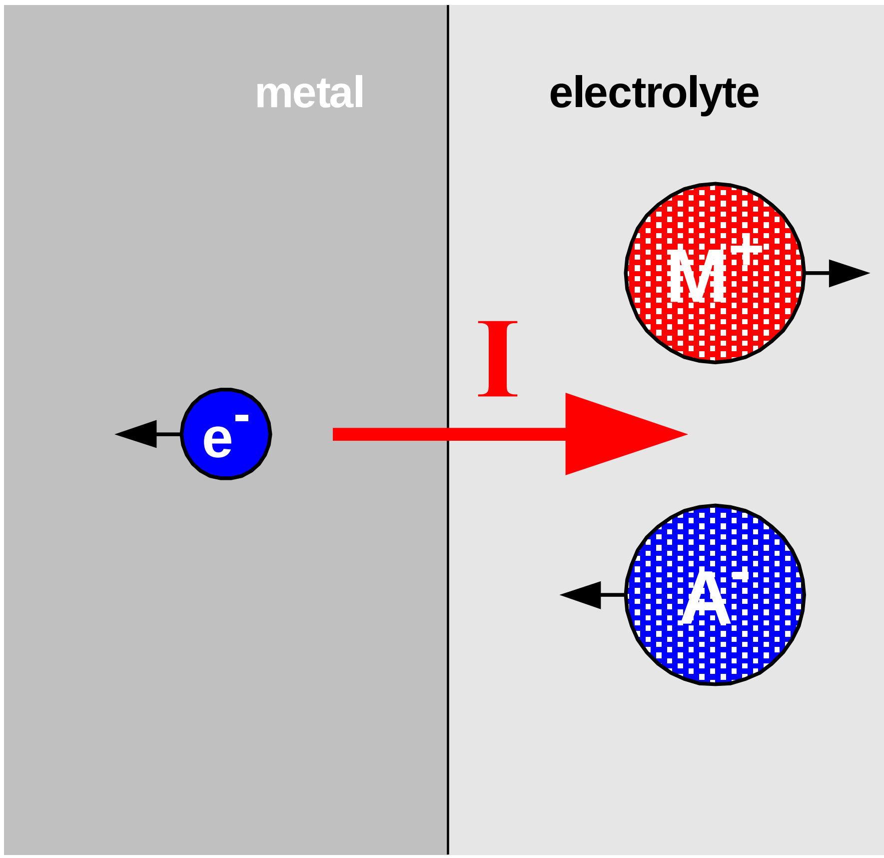
- The electrode is made up of some atoms of the same material as the cations and that this material in the electrode at the interface can become oxidized to form a cation and one or more free electrons
- The cation is discharged into the electrolyte; the electron remains as a charge carrier in the electrode.

$$A^{m-} \rightleftarrows A + me^{-}$$

and *m* is the valence of *A*

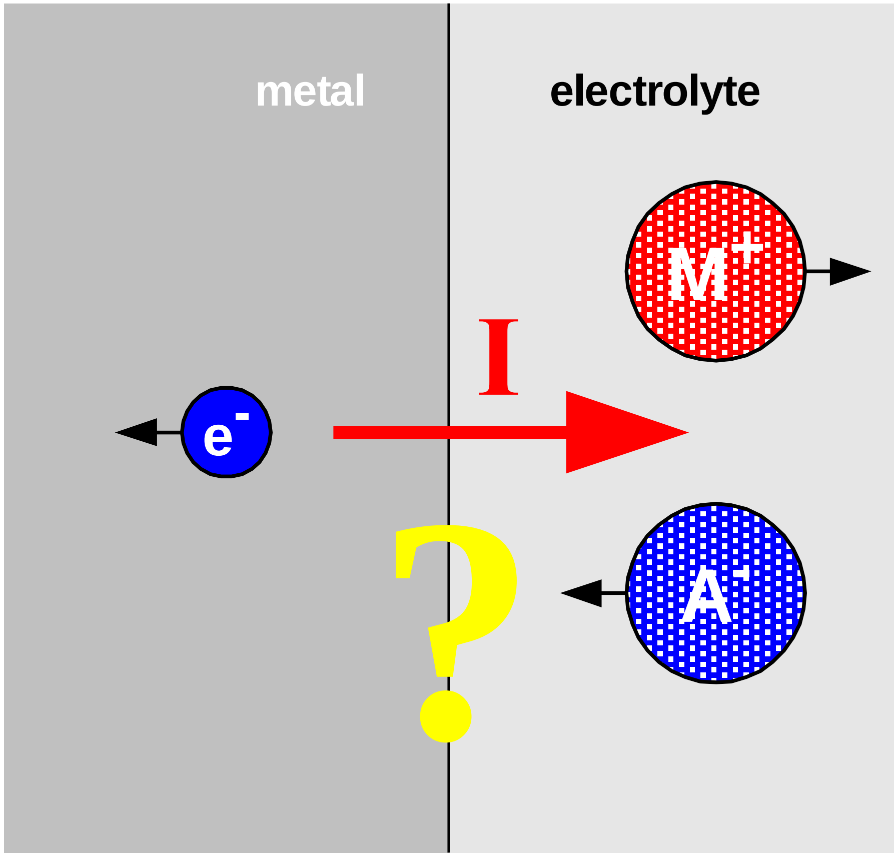
- an anion coming to the electrode–electrolyte interface can be oxidized to a neutral atom, giving off one or more free electrons to the electrode.

Electrode-Electrolyte interface



To sense a signal
a current **I must flow** !

Electrode-electrolyte Interface problem



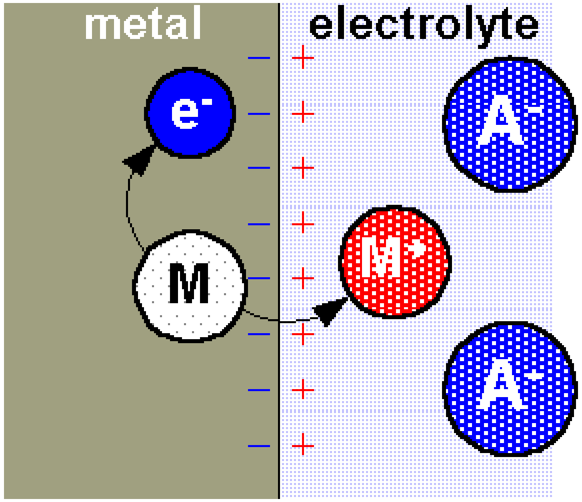
To sense a signal
a current **I must flow** !

But no electron **e$^-$** is passing the interface!

metal cation

leaving into the electrolyte

No current



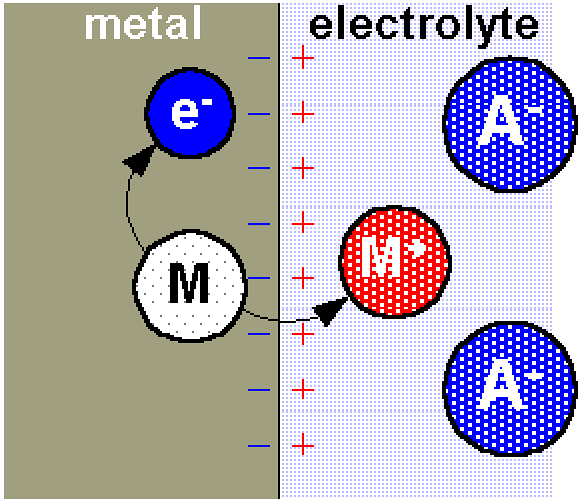
What’s going on?

www.vyssotski.ch/BasicsOfInstrumentation/Electrodes.ppt

metal cation

leaving into the electrolyte

No current



One atom M out of the metal is oxidized to form one cation M^+ and giving off one free electron e^- to the metal.

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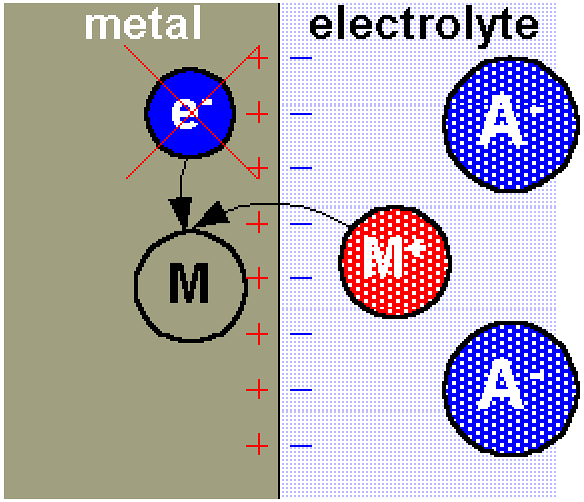
metal cation

joining the metal

No current

What's going on?

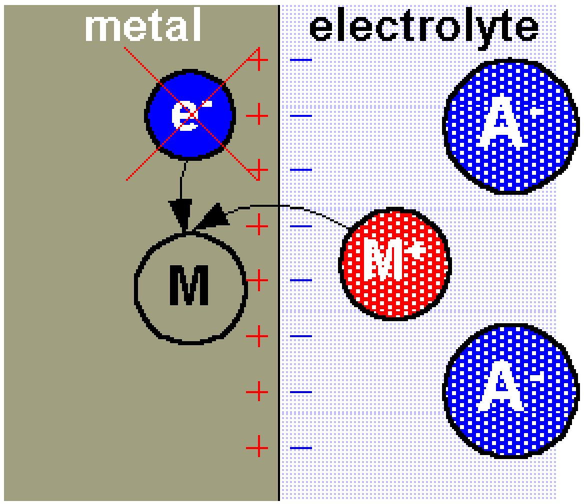


metal cation

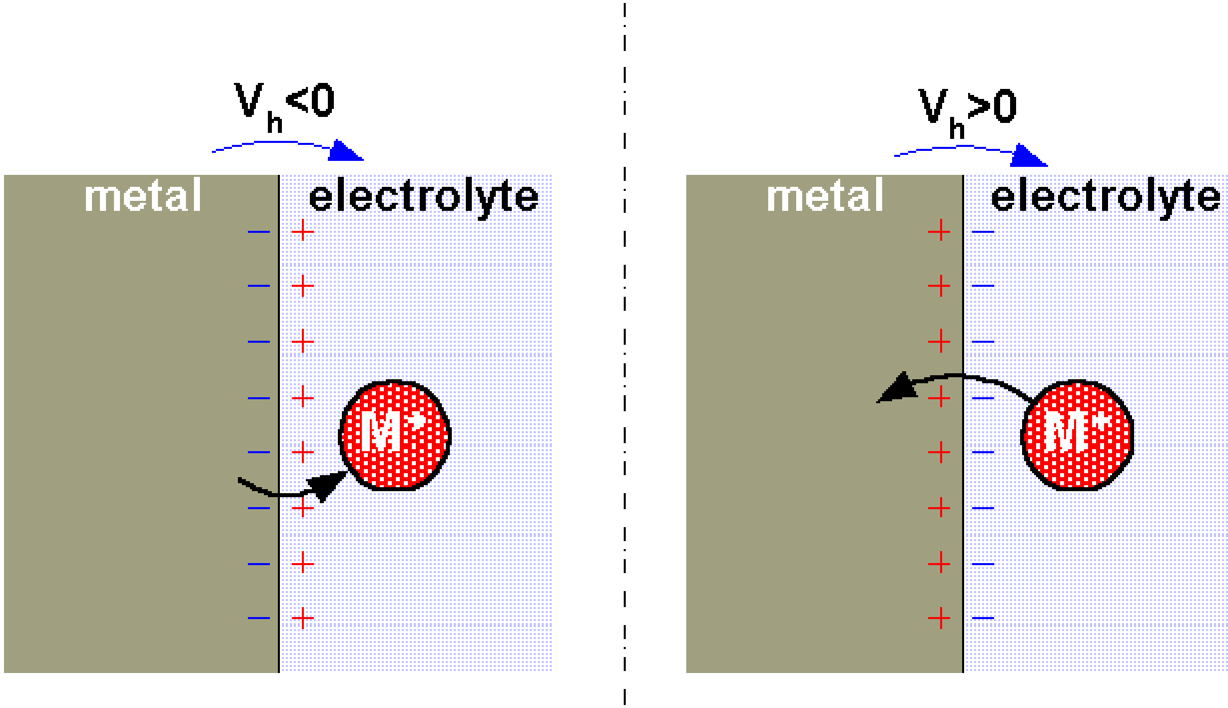
joining the metal

No current

One cation M^+
out of the electrolyte
becomes one neutral atom M
taking off one free electron
from the metal.

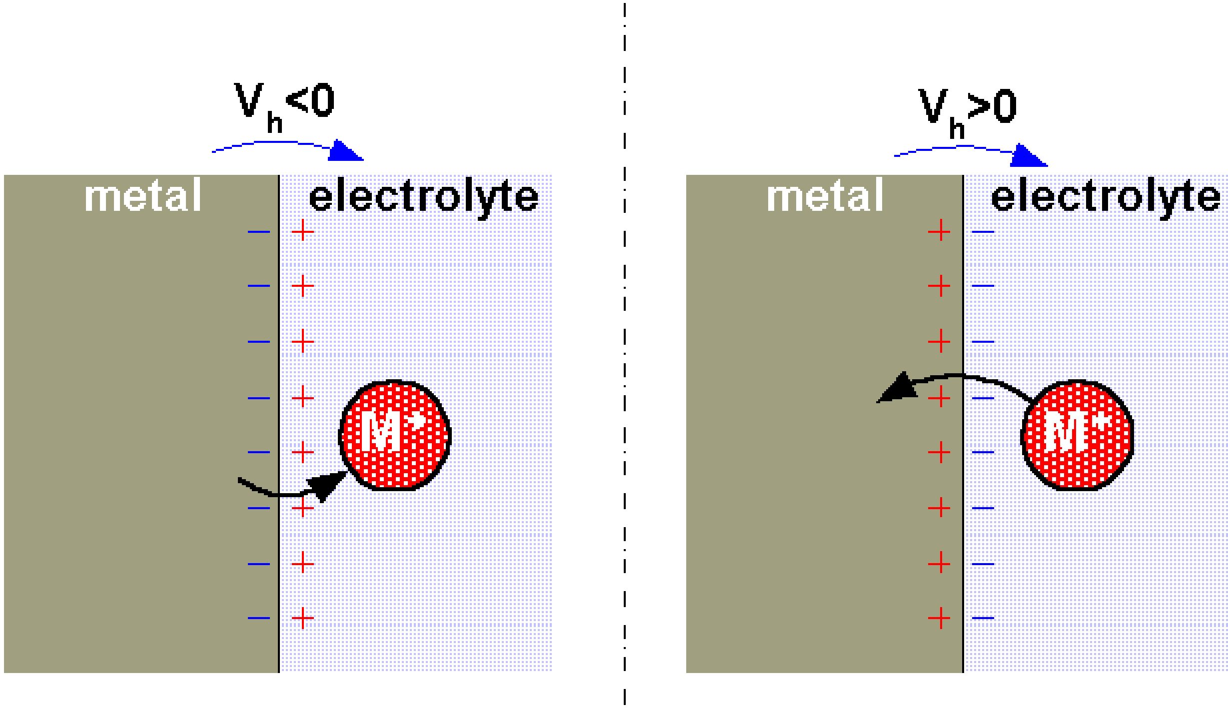


Half-cell voltage



- As reactions reach equilibrium, no current flows between the electrode and the electrolyte. so the rates of oxidation and reduction at the interface are equal.
- Under these conditions, a characteristic potential difference called **equilibrium half-cell potential** is established by the electrode and its surrounding electrolyte which depends on the metal, concentration of ions in solution and temperature (and some second order factors) .

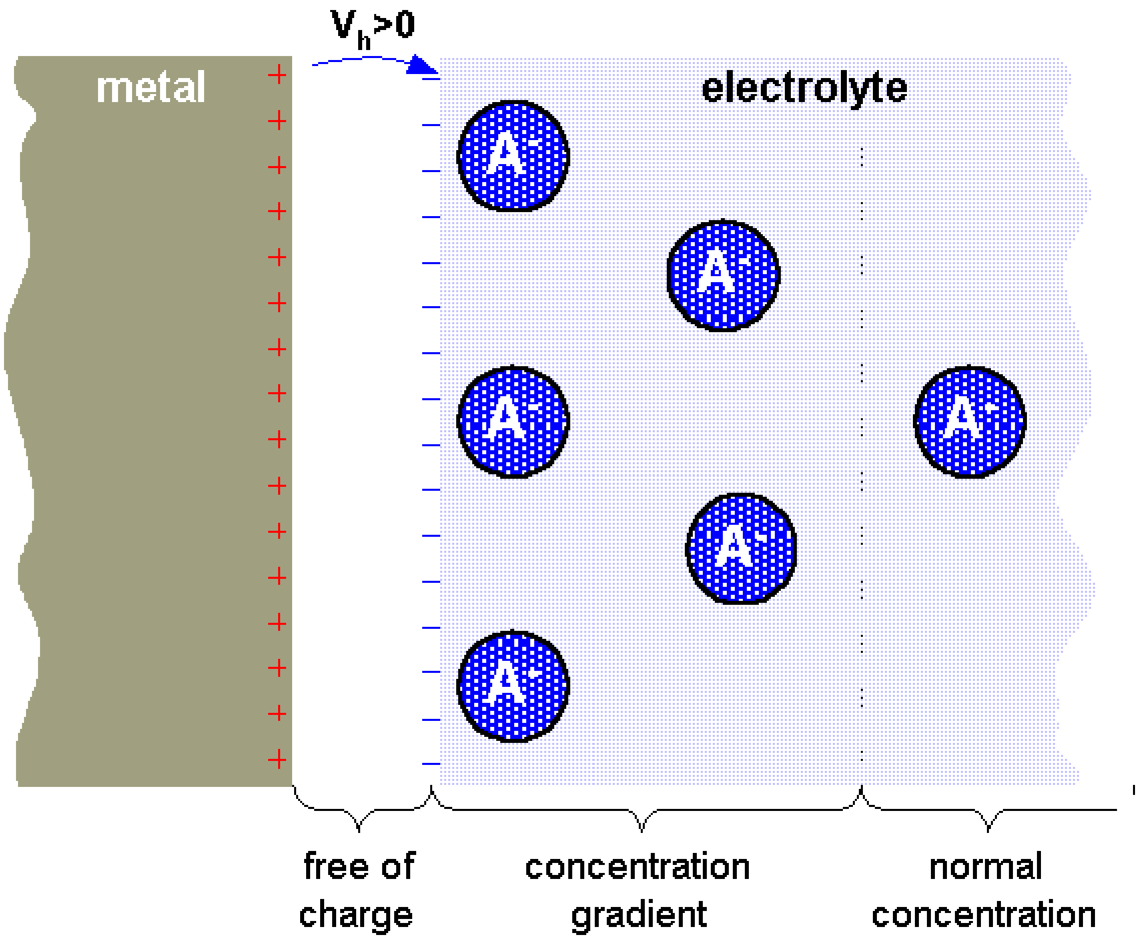
Half-cell voltage



- **Equilibrium half-cell potential** results from the distribution of ionic concentration in the vicinity of the electrode–electrolyte interface.
- Oxidation or reduction reactions at the electrode-electrolyte interface lead to a double-charge layer, similar to that which exists along electrically active biological cell membranes.
- The electrolyte surrounding the metal is at a different electric potential from the rest of the solution.

Electrode double layer

No current



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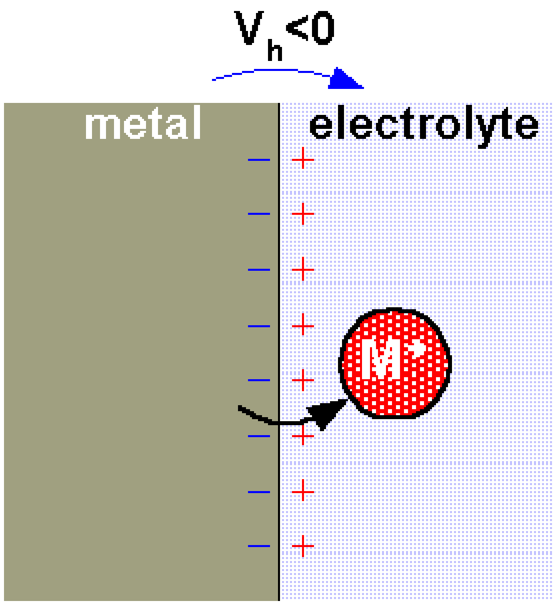
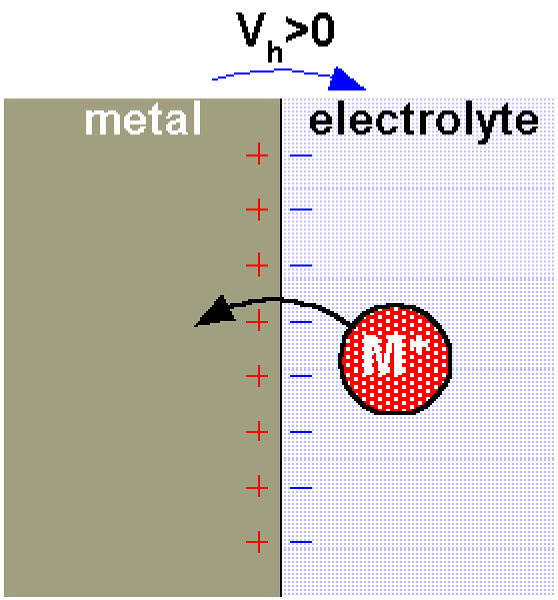
Half-cell voltage

- Half-cell potential cannot be measured without a second electrode. It is physically impossible to measure the potential of a single electrode: only the difference between the potentials of two electrodes can be measured.
- The half-cell potential of the standard hydrogen electrode has been arbitrarily set to zero. Other half cell potentials are expressed as a potential difference with this electrode.

No current, 1M salt concentration, T = 25ºC

metal:	Li	Al	Fe	Pb	H	Ag/AgCl	Cu	Ag	Pt	Au
V_h / Volt	*-3,0*	*negativ*			0	*0,223*	*positiv*			*1,68*





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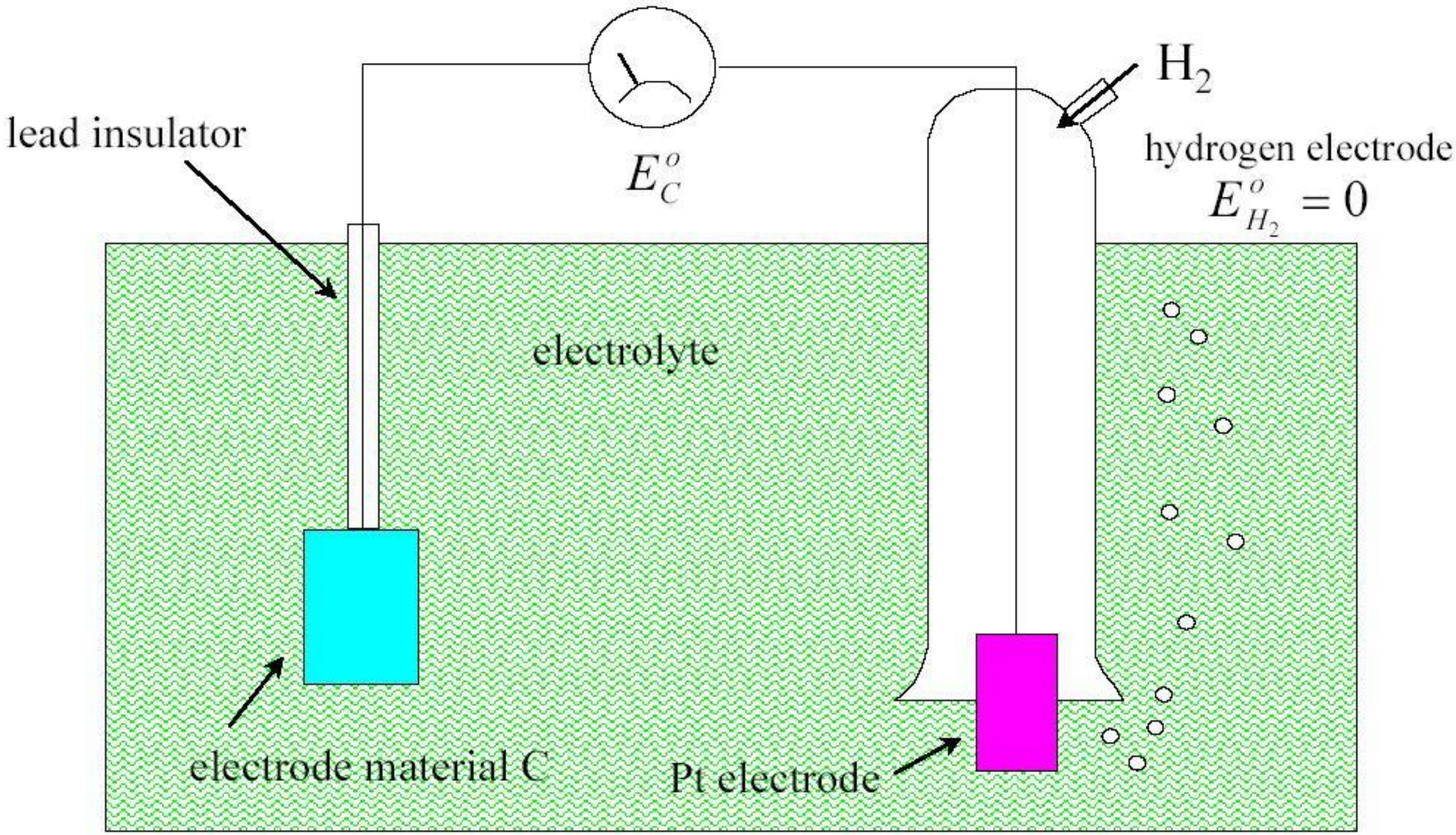
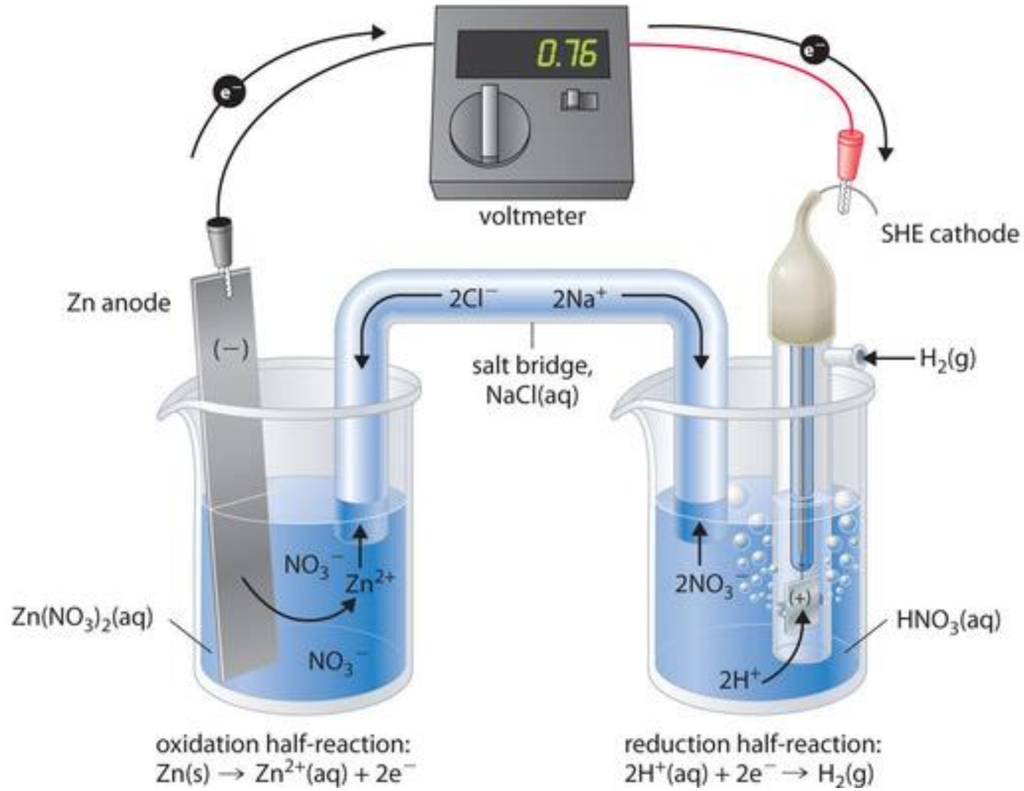
Measuring Half Cell Potential



Note: Electrode material is metal + salt or polymer selective membrane

Half Cell potential

- The standard hydrogen electrode (SHE) is universally used for reference and is assigned a standard potential of 0V.
- The $[H^+]$ in solution is in equilibrium with H_2 gas at a pressure of 1 atm at the Pt-solution interface.
- One especially attractive feature of the SHE is that the Pt metal electrode is not consumed during the reaction.



http://chemwiki.ucdavis.edu/Analytical_Chemistry/Electrochemistry/Standard_Potentials

Half Cell potential

- Cathode

$$2H^+_{(aq)} + 2e^- \rightarrow H_{2(g)} \qquad E^\circ{}_{cathode} = 0V$$

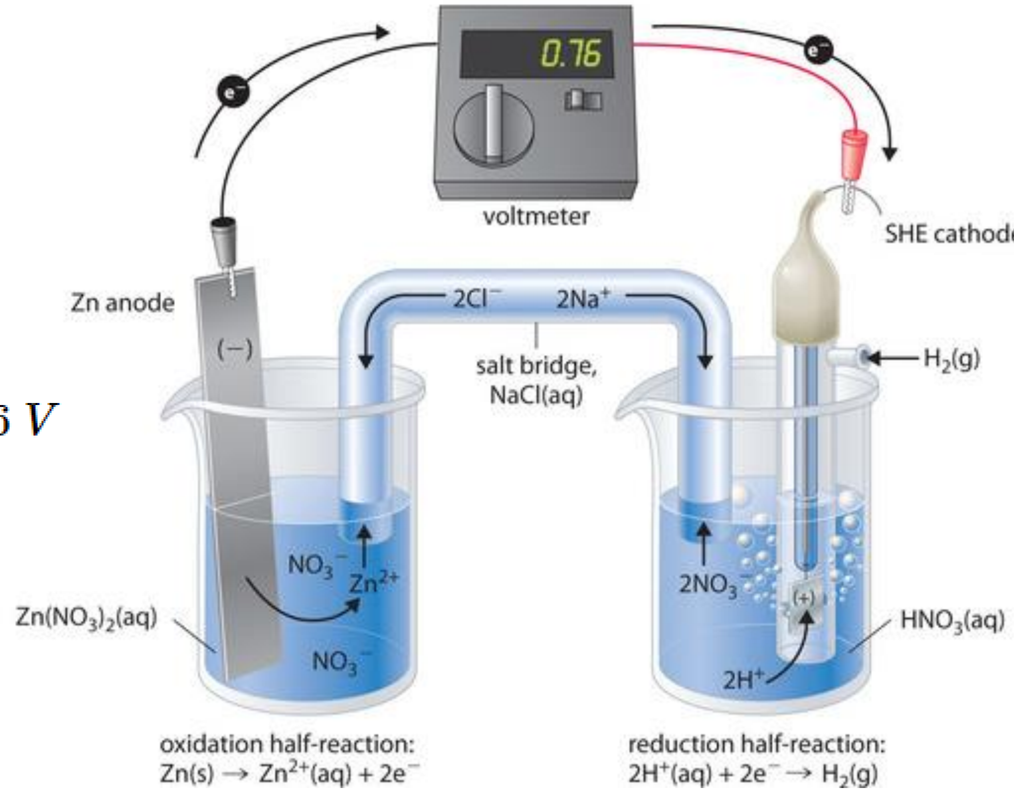
- Anode

$$Zn_{(s)} \rightarrow Zn^{2+}_{(aq)} + 2e^- \qquad E^\circ{}_{anode} = -0.76\ V$$

- Overall

$$Zn_{(s)} + 2H^+_{(aq)} \rightarrow Zn^{2+}_{(aq)} + H_{2(g)}$$

$$E^\circ{}_{cell} = E^\circ{}_{cathode} - E^\circ{}_{anode} = 0.76\ V$$



In figure there is SHE in one beaker and a Zn strip in another beaker containing a solution of Zn^{2+} ions. When the circuit is closed, the voltmeter indicates a potential of 0.76 V. The zinc electrode begins to dissolve to form Zn^{2+} , and H^+ ions are reduced to H_2 in the other compartment. Thus the hydrogen electrode is the cathode, and the zinc electrode is the anode.

http://chemwiki.ucdavis.edu/Analytical_Chemistry/Electrochemistry/Standard_Potentials

Table 5.1 Half-cell Potentials for Common Electrode Materials at 25 °C

The metal undergoing the reaction shown has the sign and potential E^0 when referenced to the hydrogen electrode

Metal and Reaction	Potential E^0 (V)
$Al \rightarrow Al^{3+} + 3e^-$	−1.706
$Zn \rightarrow Zn^{2+} + 2e^-$	−0.763
$Cr \rightarrow Cr^{3+} + 3e^-$	−0.744
$Fe \rightarrow Fe^{2+} + 2e^-$	−0.409
$Cd \rightarrow Cd^{2+} + 2e^-$	−0.401
$Ni \rightarrow Ni^{2+} + 2e^-$	−0.230
$Pb \rightarrow Pb^{2+} + 2e^-$	−0.126
Standard Hydrogen electrode → $H_2 \rightarrow 2H^+ + 2e^-$	0.000 by definition
$Ag + Cl^- \rightarrow AgCl + e^-$	+0.223
$2Hg + 2Cl^- \rightarrow Hg_2Cl_2 + 2e^-$	+0.268
$Cu \rightarrow Cu^{2+} + 2e^-$	+0.340
$Cu \rightarrow Cu^+ + e^-$	+0.522
$Ag \rightarrow Ag^+ + e^-$	+0.799
$Au \rightarrow Au^{3+} + 3e^-$	+1.420
$Au \rightarrow Au^+ + e^-$	+1.680

SOURCE: Data from *Handbook of Chemistry and Physics*, 55th ed., Cleveland, OH: CRC Press, 1974–1975, with permission.

$$H_2 \rightleftarrows 2H \rightleftarrows 2H^+ + 2e^-$$

Some half cell potentials

reduction reaction	$E^o\ (V)$
$Al^{3+} + 3e^- \rightarrow Al$	-1.662
$Zn^{2+} + 2e^- \rightarrow Zn$	-0.762
$Cr^{3+} + 3e^- \rightarrow Cr$	-0.744
$Fe^{2+} + 2e^- \rightarrow Fe$	-0.447
$Cd^{2+} + 2e^- \rightarrow Cd$	-0.403
$Ni^{2+} + 2e^- \rightarrow Ni$	-0.257
$Pb^{2+} + 2e^- \rightarrow Pb$	-0.126
$2H^+ + 2e^- \rightarrow H_2$	0.000
$AgCl + e^- \rightarrow Ag + Cl^-$	$+0.222$
$Hg_2Cl_2 + 2e^- \rightarrow 2Hg + 2Cl^-$	$+0.268$
$Cu^{2+} + 2e^- \rightarrow Cu$	$+0.342$
$Cu^+ + e^- \rightarrow Cu$	$+0.521$
$Ag^+ + e^- \rightarrow Ag$	$+0.780$
$Au^{3+} + 3e^- \rightarrow Au$	$+1.498$
$Au^+ + e^- \rightarrow Au$	$+1.692$

Standard Hydrogen electrode

Note: Ag-AgCl has low junction potential & it is also very stable -> hence used in ECG electrodes!

Polarization

If there is a current between the electrode and electrolyte, the observed half cell potential is often altered due to polarization. The difference is due to polarization of the electrode.

Overpotential
Difference between observed and zero-current half cell potentials

Resistance
Current changes resistance of electrolyte and thus, a voltage drop results.

Concentration
Changes in distribution of ions at the electrode-electrolyte interface

Activation
The activation energy barrier depends on the direction of current and determines kinetics

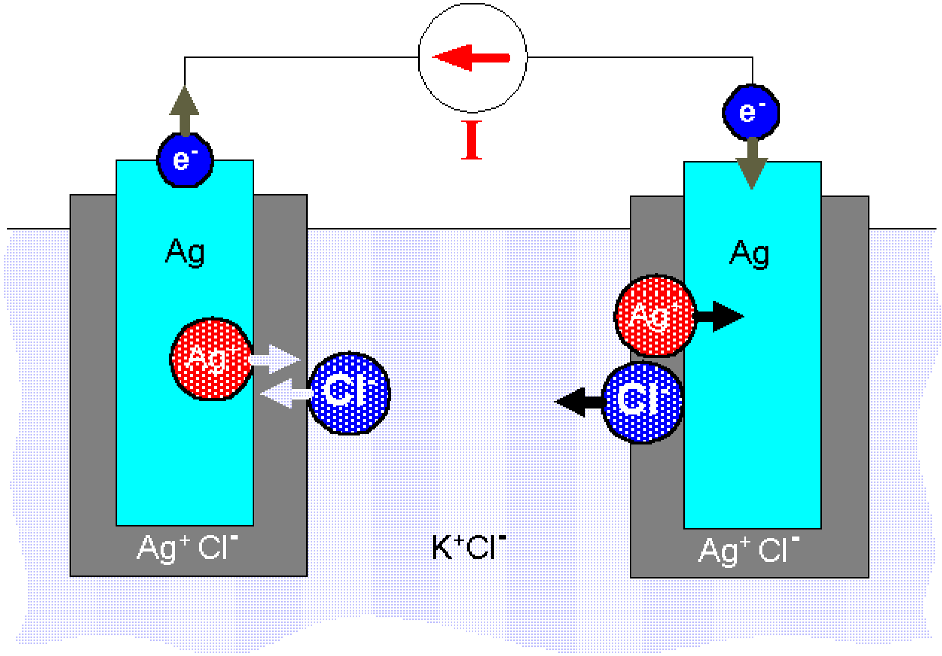
$$V_p = V_R + V_C + V_A + E_0$$

Note: Polarization and impedance of the electrode are two of the most important electrode properties to consider.

Ohmic Overpotential

- Direct result of the resistance of the electrolyte.

- When a current passes between two electrodes immersed in an electrolyte, there is a voltage drop along the path of the current in the electrolyte as a result of its resistance.

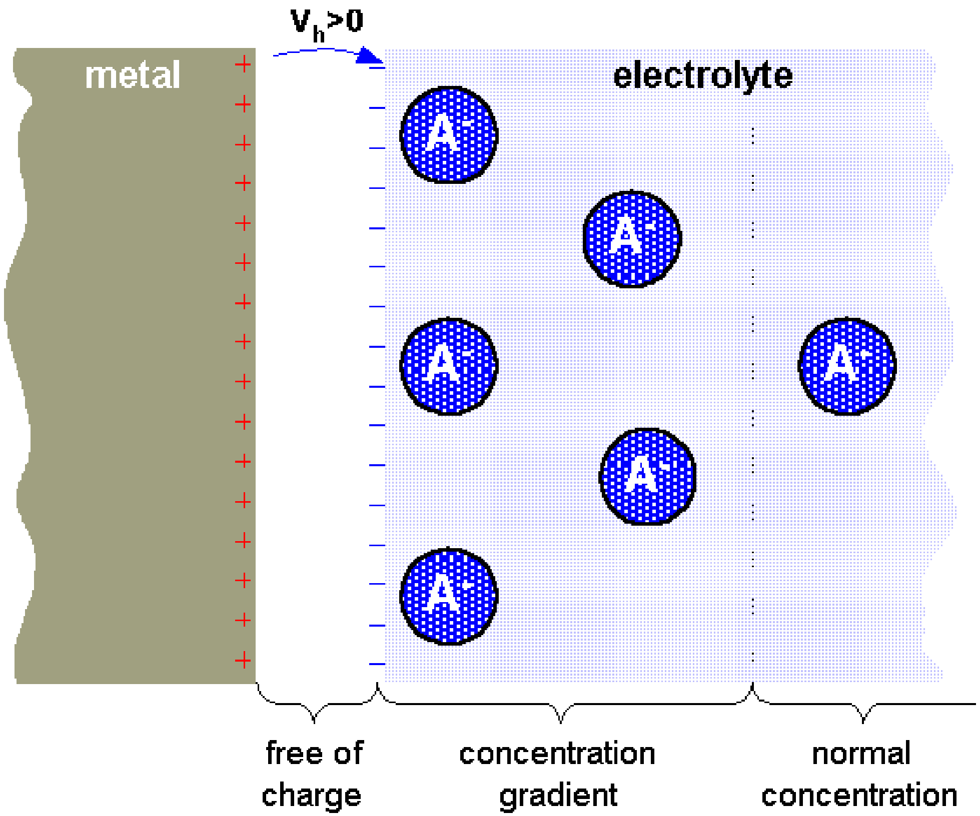


- This drop in voltage is proportional to the current and the resistivity of the electrolyte.

Concentration Overpotential

- Results from changes in the distribution of ions in the electrolyte in the vicinity of the electrode–electrolyte interface.

- When a current is established, the rates of oxidation and reduction at the interface are no longer equal.
- Thus it is reasonable to expect the concentration of ions to change.



- This change results in a different half-cell potential at the electrode. The difference between this and the equilibrium half-cell potential is the concentration overpotential.

Nernst Equation

- When two aqueous ionic solutions of different concentration are separated by an ion-selective semipermeable membrane, an electric potential exists across this membrane.
- Nernst equation

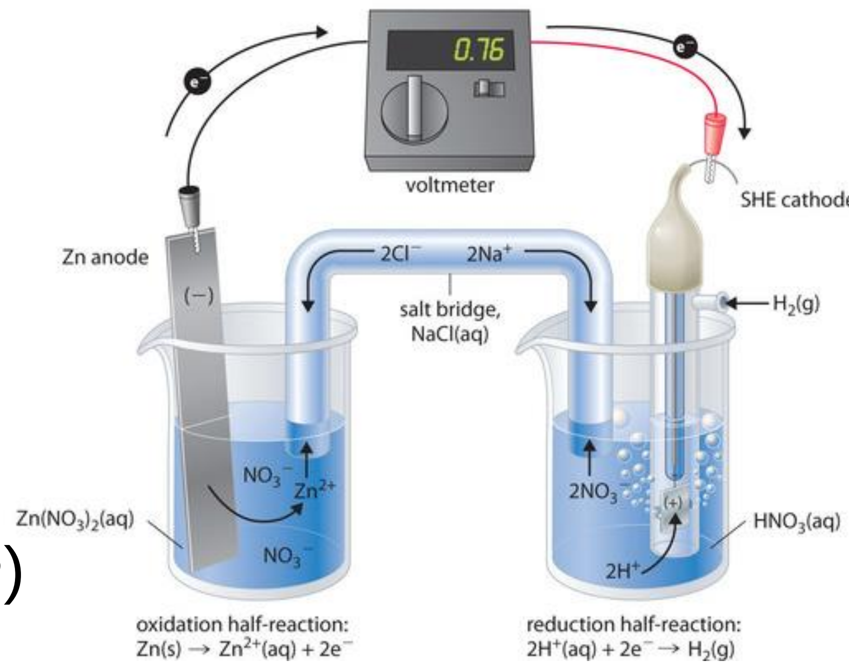
$$E = -\frac{RT}{nF}\ln\frac{a_1}{a_2}$$

where a_1 and a_2 are the activities of the ions on each side of the membrane.

Half Cell potential

- When the electrode–electrolyte system no longer maintains this standard condition, half-cell potentials different from the standard half-cell potential are observed.
- The differences in potential are determined primarily by temperature
- and ionic activity in the electrolyte

$$E = E^0 + \frac{RT}{nF}\ln(a_{c^{n+}})$$



E = half-cell potential
E^0 = standard half-cell potential
n = valence of electrode material
a_c^{n+} = activity of cation C^{n+}
(*generally same as concentration*)

General Nernst Equation

- The more general form of this equation can be written for a general oxidation–reduction reaction as

$$\alpha A + \beta B \rightleftarrows \gamma C + \delta D + n\mathrm{e}^{-}$$

where *n* electrons are transferred.

The general Nernst equation for this situation is

$$E = E^{0} = \frac{RT}{nF} \ln \frac{a_C^{\gamma} a_D^{\delta}}{a_A^{\alpha} a_B^{\beta}}$$

where the *a*'s represent the activities of the various constituents of the reaction.

Activation Overpotential

- The charge-transfer processes involved in the oxidation–reduction reaction are not entirely reversible.
- In order for metal atoms to be oxidized to metal ions that are capable of going into solution, the atoms must overcome an energy barrier which governs the kinetics of the reaction.
- The reverse reaction—in which a cation is reduced, thereby plating out an atom of the metal on the electrode—also involves an activation energy, but it does not necessarily have to be the same as that required for the oxidation reaction.
- When there is a current between the electrode and the electrolyte, either oxidation or reduction predominates, and hence the height of the energy barrier depends on the direction of the current.
- This difference in energy appears as a difference in voltage between the electrode and the electrolyte, which is known as the *activation overpotential.*

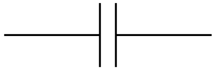
- Note: for a metal electrode, 2 processes can occur at the electrolyte interfaces:

– A capacitive process resulting from the redistribution of charged and polar particles with nocharge-transfer between the solution and the electrode

– A component resulting from the electron exchange between the electrode and a redox species in the solution termed *faradaic process.*

Polarizable and Non-Polarizable Electrodes

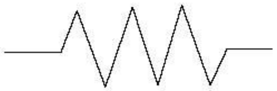
<u>Perfectly Polarizable Electrodes</u>



Use for recording

These are electrodes in which **no crosses the electrode-electrolyte interface** when a current is applied. The current across the interface is a **displacement current** and the electrode behaves like a capacitor.
Example: Platinum Electrode (Noble metal)

<u>Perfectly Non-Polarizable Electrode</u>



Use for stimulation

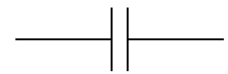
These are electrodes where **current passes freely across the electrode-electrolyte interface**, requiring no energy to make the transition. These electrodes see **no overpotentials**.
Example : Ag/AgCl electrode

Example: Ag-AgCl is used in recording while Pt is use in stimulation

Polarizable and Non-Polarizable Electrodes

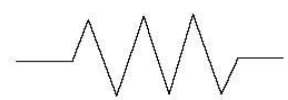
No electrode is perfectly polarizable or non-polarizable.

Platinum electrodes are a reasonable approximation of **perfectly polarizable electrodes.**



- They exhibit Vp that primarily results from Vc and Va.
- Used for stimulation and for higher frequency biopotential measurements.

Ag/AgCl electrodes behave reasonably closely to **perfectly non-polarizable electrodes.**



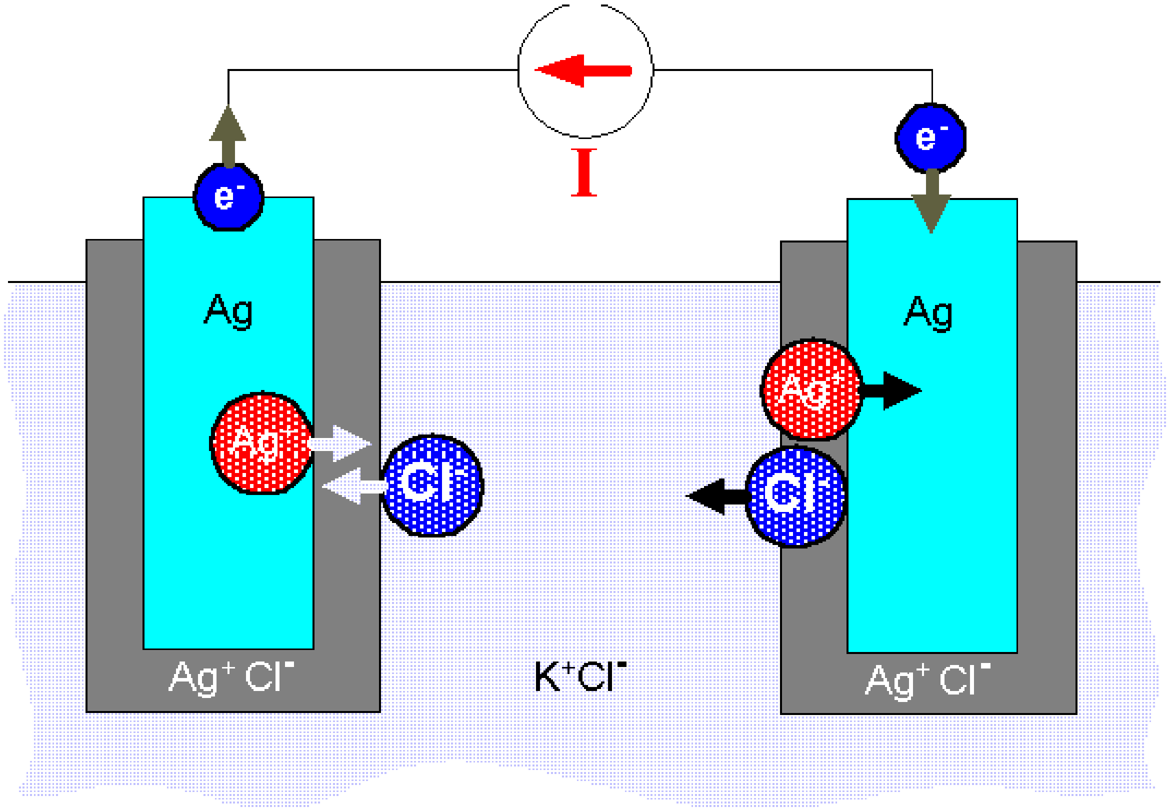
- They exhibit Vp that primarily results from Vr only.
- Used for biopotential recordings that range from high frequency to very low frequency.

Motion artifact

- If a pair of electrodes is in an electrolyte and one moves with respect to the other, a potential difference appears across the electrodes known as the ***motion artifact.*** This is a source of noise and interference in bio-potential measurements.
- When the electrode moves with respect to the electrolyte, the distribution of the double layer of charge on polarizable electrode interface changes. This changes the half-cell potential temporarily.
- **Note:** Motion artifact is minimal for non-polarizable electrodes (Measurement electrodes – AgCl).

silver/silver chloride electrodes

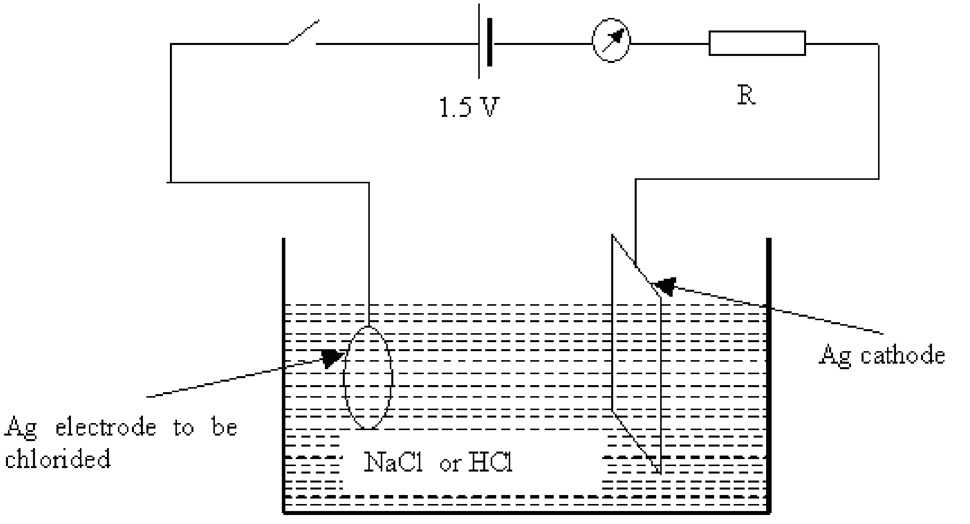
- Ag/AgCl is a practical electrode that approaches the characteristics of a perfectly nonpolarizable electrode
- Can be easily fabricated in the laboratory.



- Consists of a metal coated with a layer of a slightly soluble ionic compound of that metal with a suitable anion.
- Whole structure is immersed in an electrolyte containing the anion in relatively high concentrations.

Electrolytic process - Ag/AgCl electrode fabrication

- An electrochemical cell with the Ag electrode on which the AgCl layer is to be deposited serves as anode and another piece of Ag—having a surface area much greater than that of the anode—serves as cathode.



- The reactions begin to occur as soon as the battery is connected, and the current jumps to its maximal value.
- As the thickness of the deposited AgCl layer increases, the rate of reaction decreases and the current drops.
- This situation continues, and the current approaches zero asymptotically.

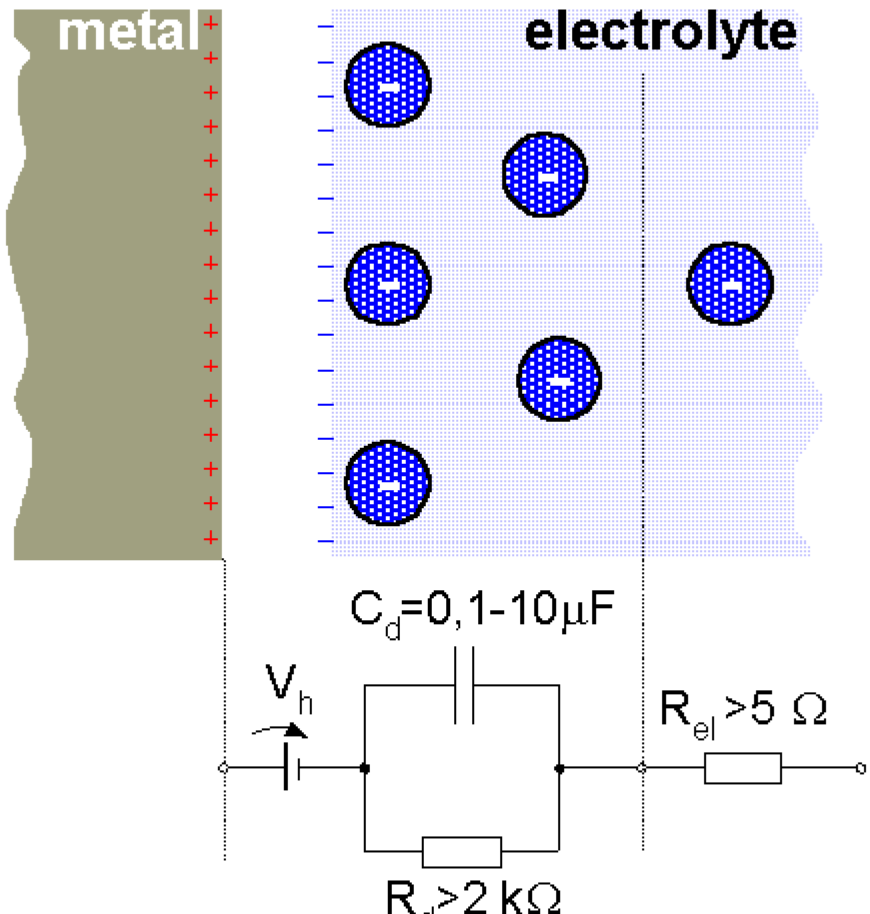
faculty.ksu.edu.sa/ElBrawany/Courses/Med201Files/W7.ppt

Electrode behavior and circuit model

- Current–voltage characteristics of the electrode–electrolyte interface are found to be nonlinear,
- Nonlinear elements are required for modeling electrode behavior.
- Characteristics of an electrode are sensitive to the current passing through the electrode,
- Electrode characteristics at relatively high current densities can be considerably different from those at low current densities.
- Characteristics of electrodes are also waveform-dependent.
- When sinusoidal currents are used to measure the electrode’s circuit behavior, the characteristics are also frequency-dependent.

Electrode behavior and circuit model

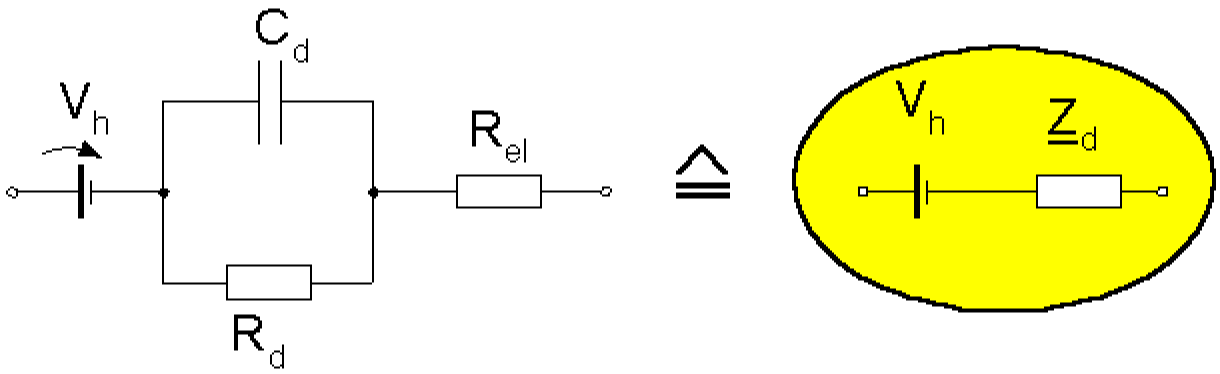
Rel=*R*s is the series resistance associated with interface effects and due to resistance in the electrolyte.



equivalent circuit

equivalent circuit

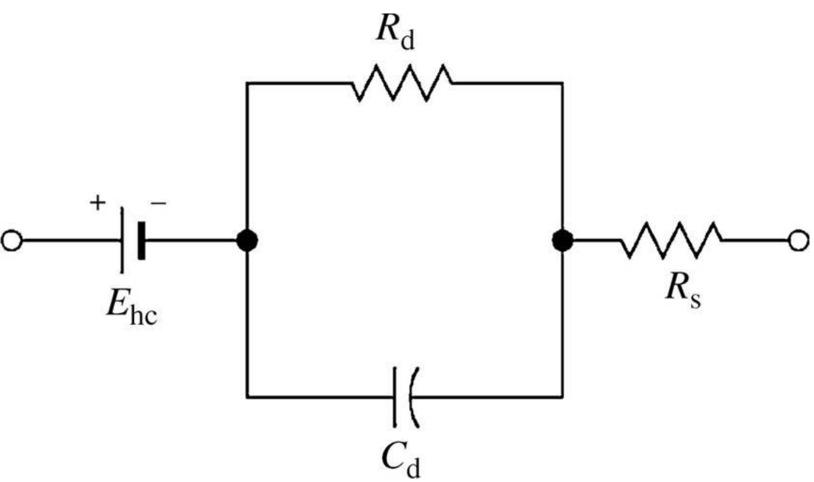
electrode-electrolyte



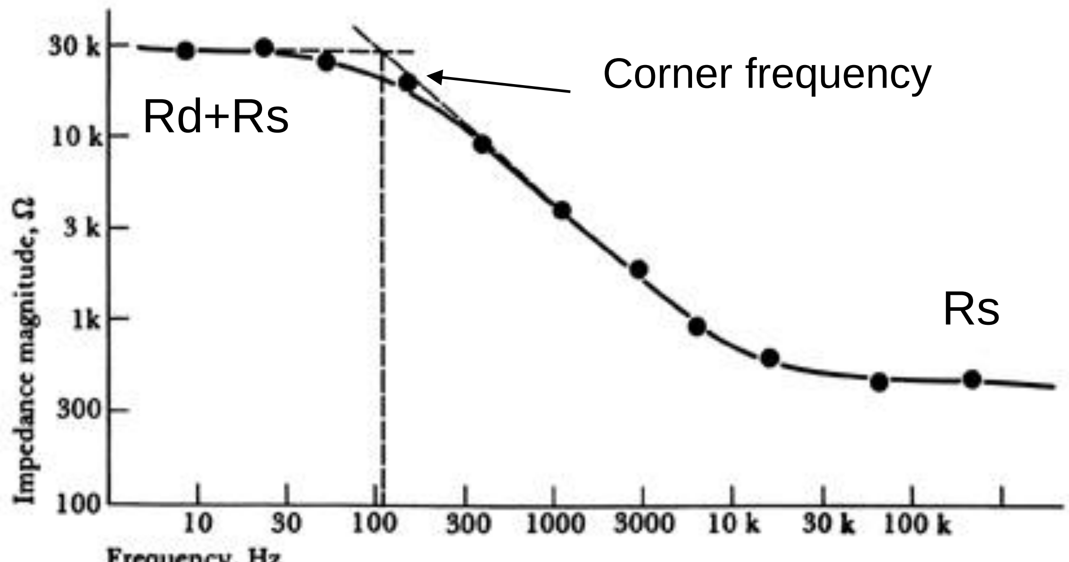
- Circuit values determined by the electrode material and its geometry, and—to a lesser extent—by the material of the electrolyte and its concentration.

Electrode impedance is frequency-dependent

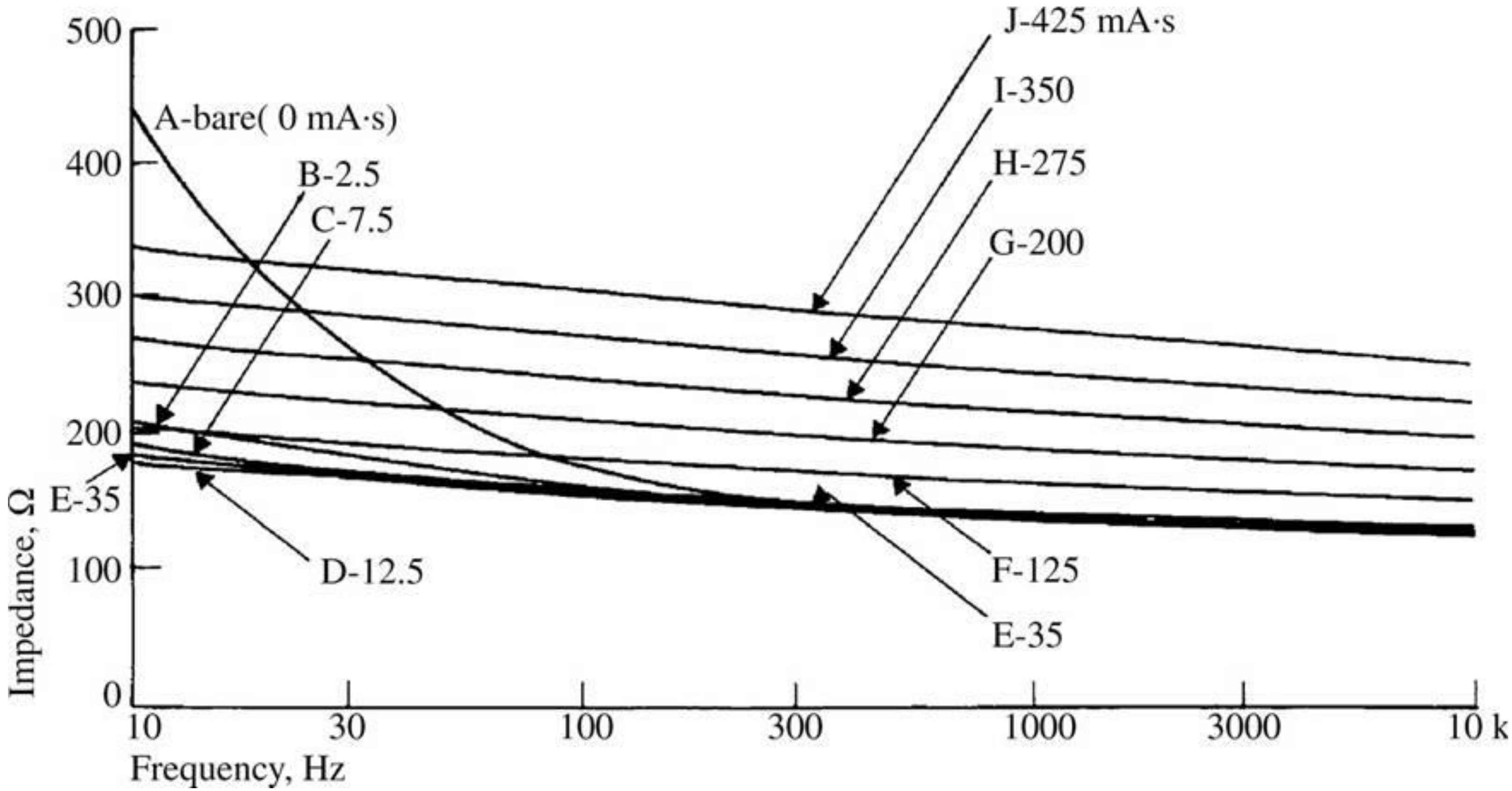
- At high frequencies, where $1/wC \ll R_d$, the impedance is constant at R_s.
- At low frequencies, where $1/wC \gg R_d$, the impedance is again constant but its value is larger, being $R_s + R_d$.
- At frequencies between these extremes, the electrode impedance is frequency-dependent.



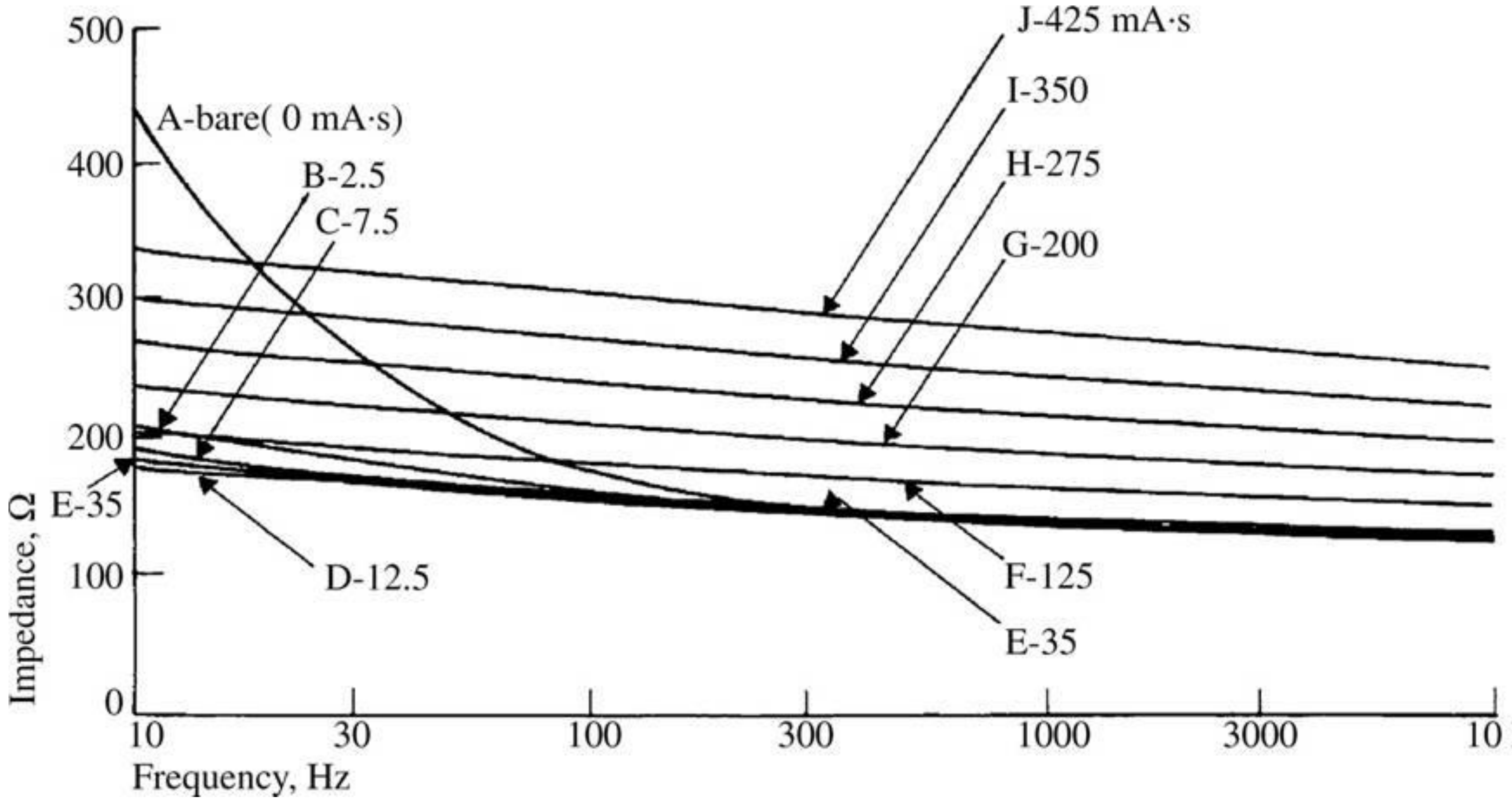
Frequency Response



- For 1 cm^2 at 10 Hz, a nickel- and carbon-loaded silicone rubber electrode has an impedance of approximately 30 kΩ,
- Whereas Ag/AgCl has an impedance of less than 10Ω.



- A metallic silver electrode having a surface area of 0.25 cm^2 had the impedance characteristic shown by curve A.
- Numbers attached to curves indicate number mA.s for each deposit.
- At a frequency of 10 Hz, the magnitude of its impedance was almost three times the value at 300 Hz.
- This indicates a strong capacitive component to the equivalent circuit.



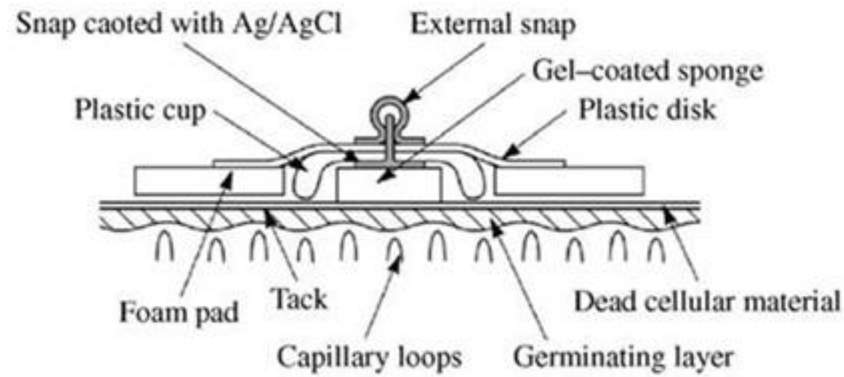
- Electrolytically depositing 2.5 mA·s of AgCl greatly reduced the low-frequency impedance, as reflected in curve B.
- Depositing thicker AgCl layers had minimal effects until the charge deposited exceeded approximately 100 mA·s.
- The curves were then seen to shift to higher impedances in a parallel fashion as the amount of AgCl deposited increased.

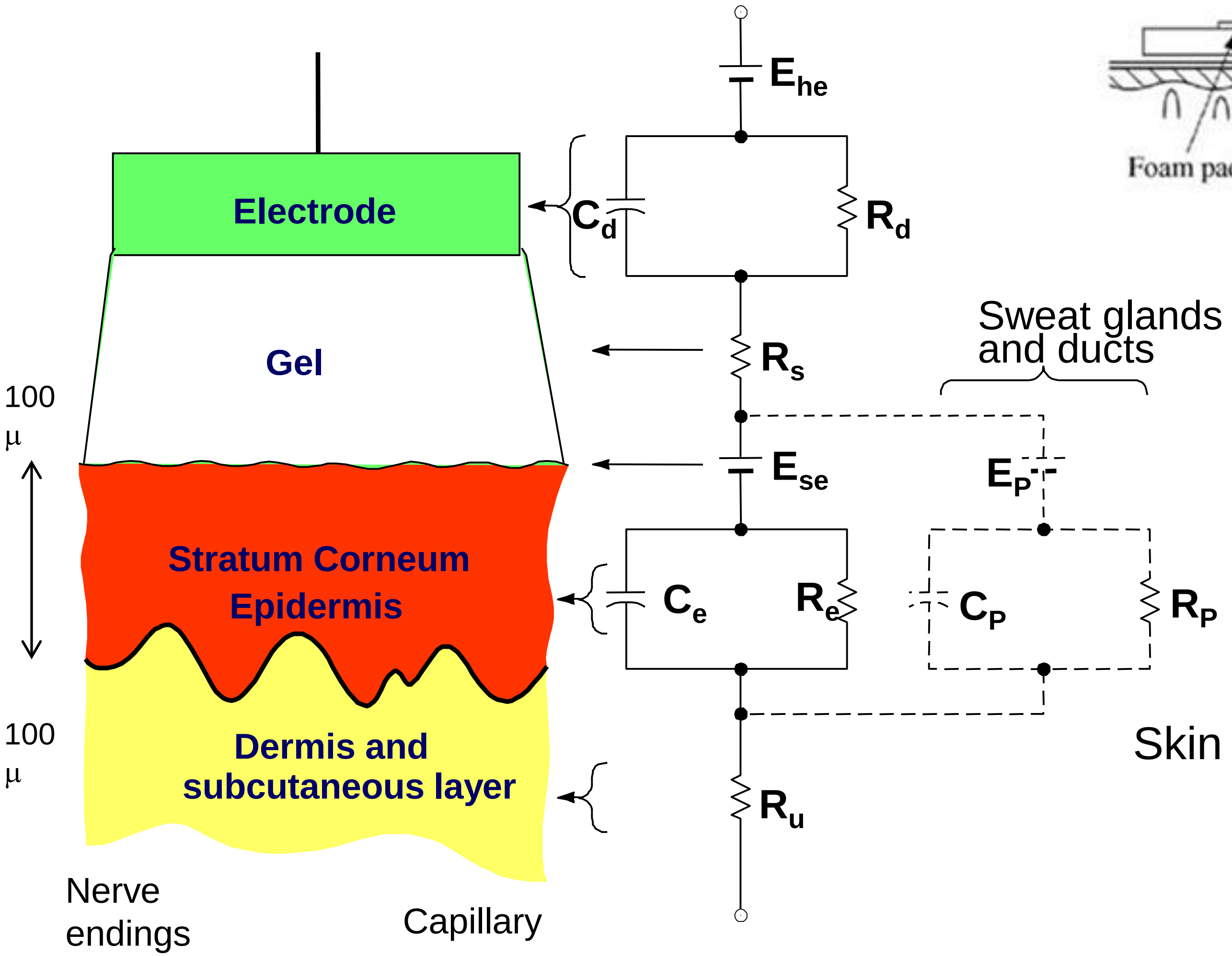
(From L. A. Geddes, L. E. Baker, and A. G. Moore, "Optimum Electrolytic Chloriding of Silver Electrodes," ***Medical and Biological Engineering,*** **1969, 7, pp. 49–56.)**

Electrode-skin interface

- Interface between the electrode–electrolyte and the skin needs to be considered to understand the behavior of the electrodes.
- In coupling an electrode to the skin, we generally use transparent electrolyte gel containing Cl^- as the principal anion to maintain good contact.
- The interface between this gel and the electrode is an electrode–electrolyte interface.
- However, the interface between the electrolyte and the skin is different.

Electrode Skin Interface



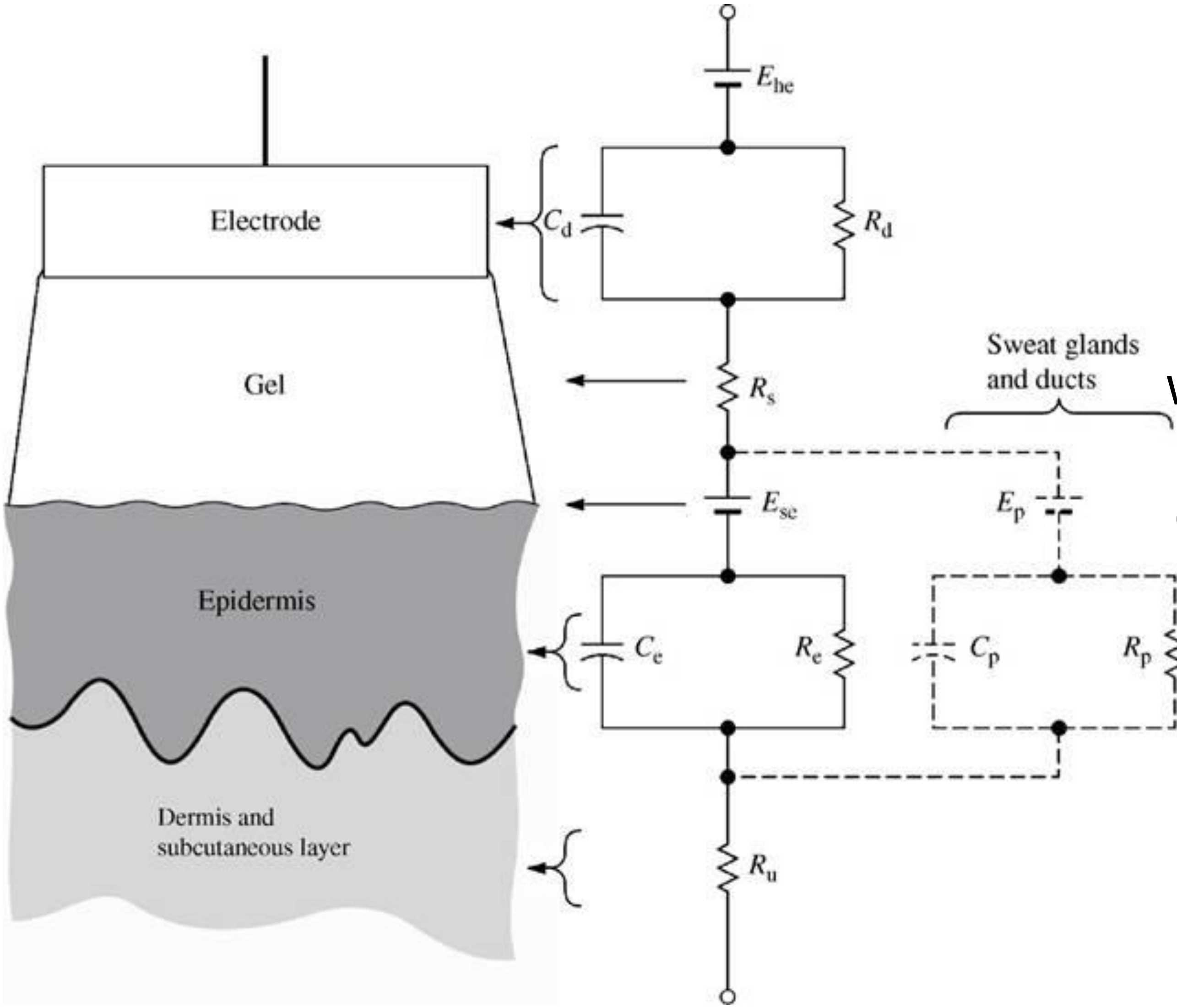


Skin impedance for 1cm^2 patch:

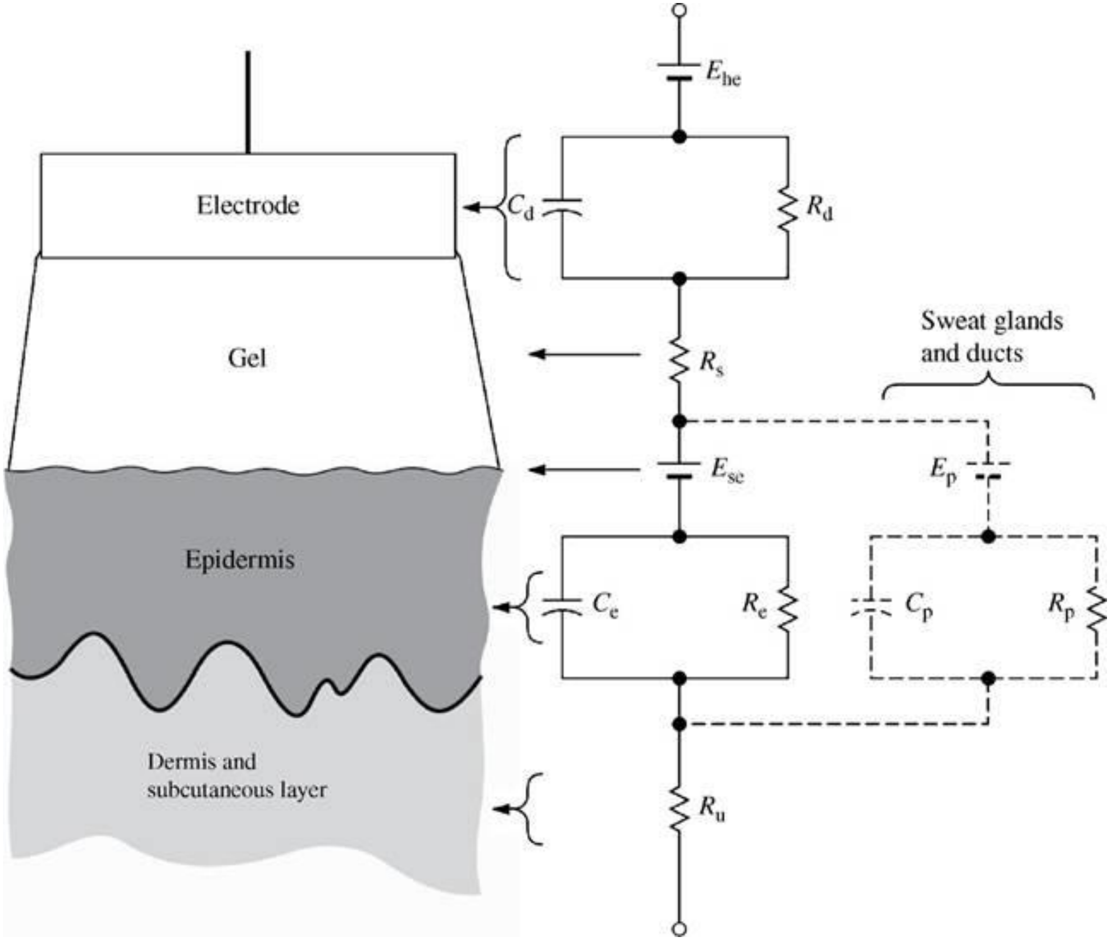
200kΩ @1Hz

200 Ω @ 1MHz

Body-surface electrode is placed against skin, showing the total electrical equivalent circuit



The series resistance R_s is now the effective resistance associated with interface effects of the gel between the electrode and the skin.



- Considering stratum corneum, as a membrane that is semipermeable to ions,
- so if there is a difference in ionic concentration across this membrane,
- there is a potential difference E_{se}, which is given by the Nernst equation.

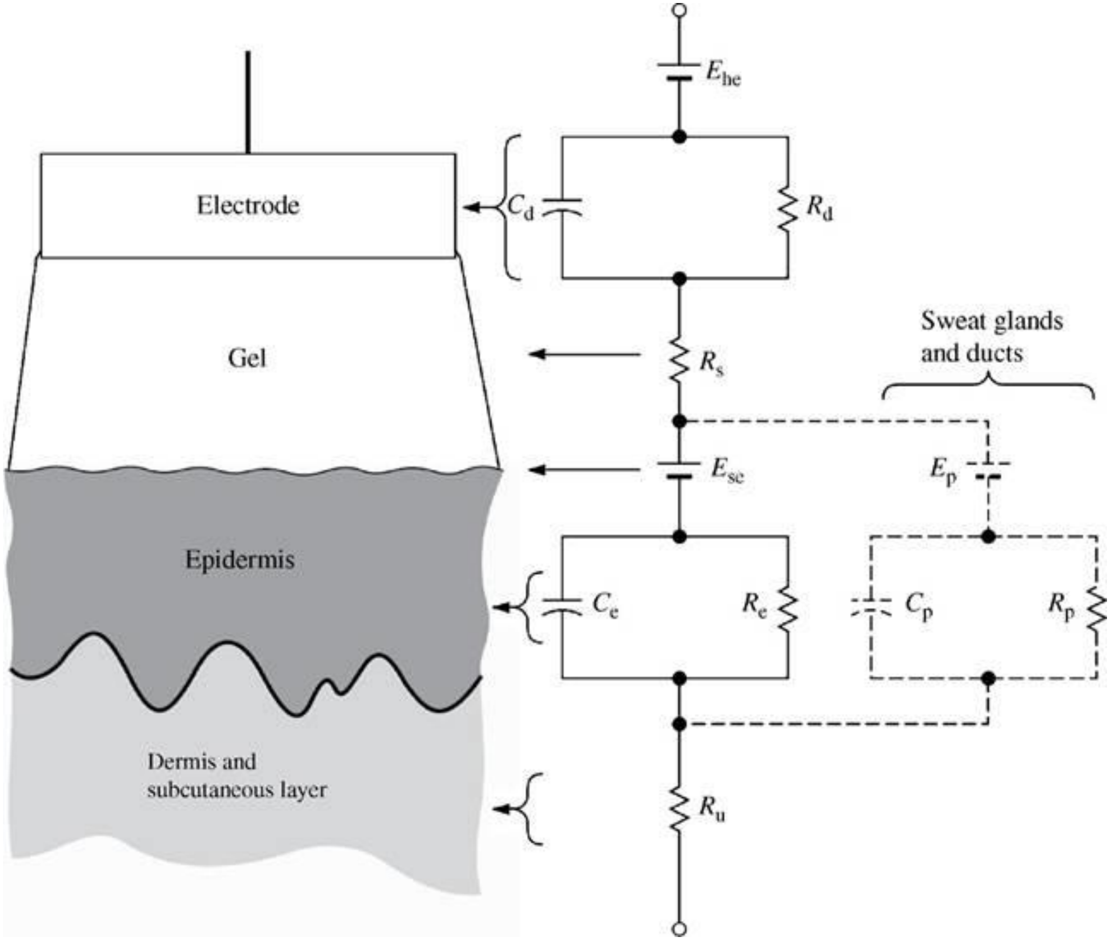
- Epidermal layer has an electric impedance that behaves as a parallel *RC* circuit.
- For 1 cm^2, skin impedance reduces from approximately 200 kΩ at 1 Hz to 200 Ω at 1 MHz.
- The dermis and the subcutaneous layer under it behave in general as pure resistances. They generate negligible dc potentials.

Reducing the effect of stratum corneum

- Minimize the effect of the stratum corneum by removing it, or at least a part of it, from under the electrode.
- Various ways:
  - vigorous rubbing with a pad soaked in acetone
  - abrading the stratum corneum with sandpaper to puncture it.
- This process tends to short out E_{se}, C_e, and R_e
- Improve stability of the signal,
  - but the stratum corneum can regenerate in as short a time as 24 h.

Psychogenic electrodermal responses or galvanic skin reflex (GSR)

- Contribution of the sweat glands and sweat ducts.
- The fluid secreted by sweat glands contains Na^+, K^+, and Cl^- ions, the concentrations of which differ from those in the extracellular fluid.
- There is a potential difference between the lumen of the sweat duct and the dermis and subcutaneous layers.



- There also is a parallel R_pC_p combination in series with this potential that represents the wall of the sweat gland and duct, as shown by the broken lines.

- These components are considered when the electrodes are used to measure the electrodermal response or GSR
- These components are often neglected when we consider biopotential electrodes

What happens when the gel dries?

- Electrolyte gel is used for good contact between electrode and skin during ECG (biopotential) measurements.
- What is the equivalent circuit when the gel is fresh?
- For prolong ambulatory recording, the gel can dry. How would be equivalent circuit change?
- How would be equivalent circuit change if the gel completely dries?
- And how would this affect the ECG recording?

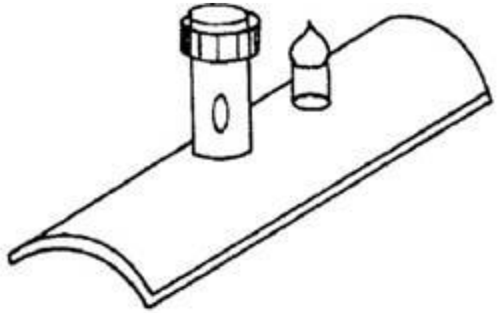
Motion Artifact

- In addition to the half-cell potential E_{hc}, the electrolyte gel–skin potential E_{se} can also cause motion artifact if it varies with movement of the electrode.
- Variations of this potential indeed do represent a major source of motion artifact in Ag/AgCl skin electrodes
- This artifact can be significantly reduced when the stratum corneum is removed by mechanical abrasion with a fine abrasive paper.
- This method also helps to reduce the epidermal component of the skin impedance.
- However, that removal of the body’s outer protective barrier makes that region of skin more susceptible to irritation from the electrolyte gel.
- Stretching the skin changes this skin potential by 5 to 10 mV, and this change appears as motion artifact

Body Surface Electrodes

- The earliest bioelectric potential measurements relied on immersion electrodes that were simply buckets of saline solution into which the patient placed a hand and a foot, as shown in Figure.
- Plate electrodes, first introduced in 1917, were a great improvement on immersion electrodes. They were originally separated from the skin by cotton pads soaked in saline to emulate the immersion electrode mechanism.
- Later, an electrolytic paste was used in place of the pad with the metal in contact with the skin.
- Electrodes that can be placed on the body surface for recording bioelectric signals.
- The integrity of the skin is not compromised.
- Can be used for short or long duration applications

1. **Metal Plate Electrodes**
2. **Suction Electrodes**
3. **Floating Electrodes**
4. **Flexible Electrodes**



(a)

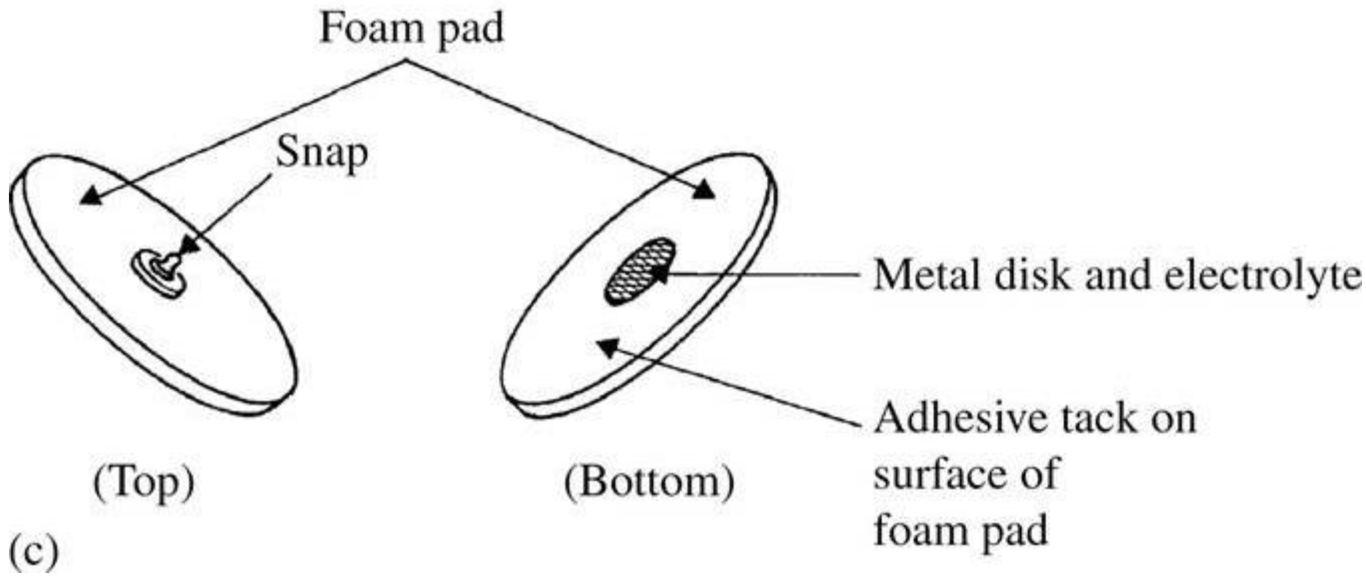
Metal-plate electrode used for application to limbs, traditionally made from German-silver (nickel-silver alloy)

(b)

Metal-disk electrode applied with surgical tape.

- Lead wire soldered or welded on the back surface.
- For ECG application made form disk of Ag with electrolytic deposition of AgCl
- For surface EMG applications made of stainless steel, platinum or gold plated disks to minimize electrolyte chemical reaction.
- Acts as polarizable electrode, and prone to motion artifacts

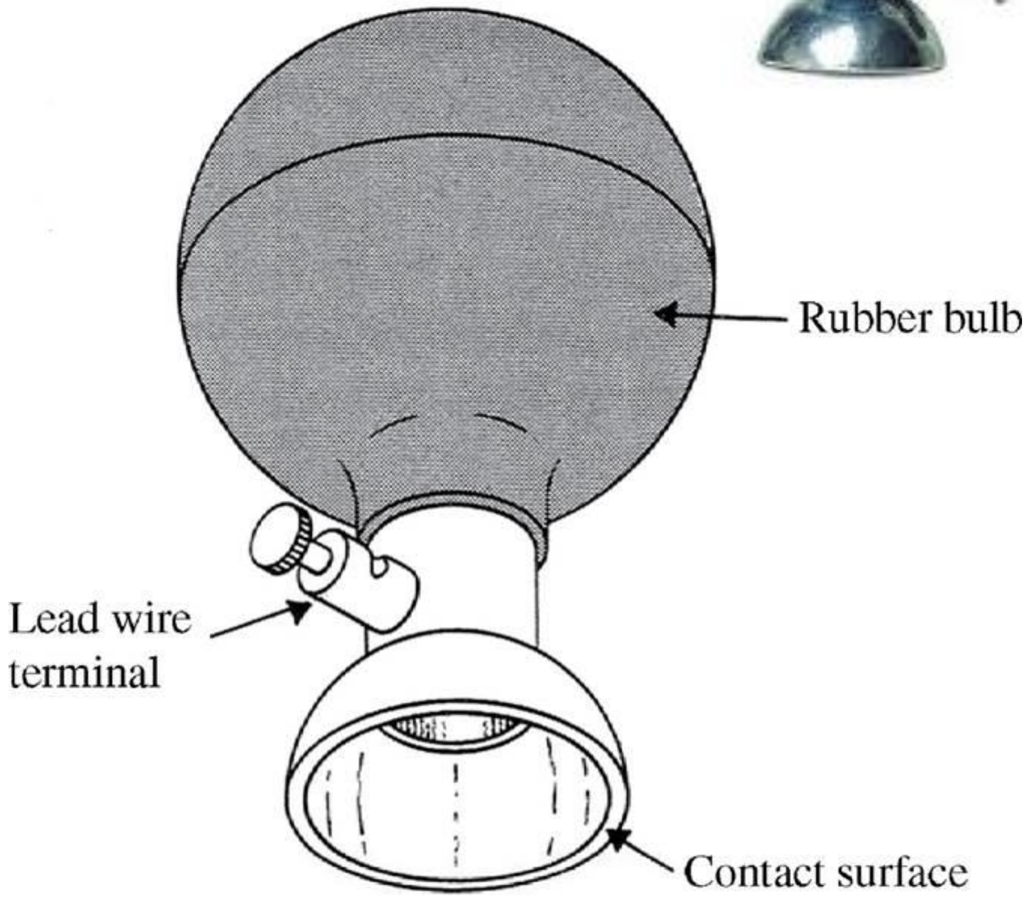
Disposable foam-pad electrodes, often used with ECG monitoring apparatus



- Relatively large disk of plastic foam with silver-plated disk serving as electrode, coated with AgCl
- Layer of electrolyte gel covers the disk
- Electrode side of foam covered with adhesive material
- They are preffered in hospitals due to being easy to apply and disposable.

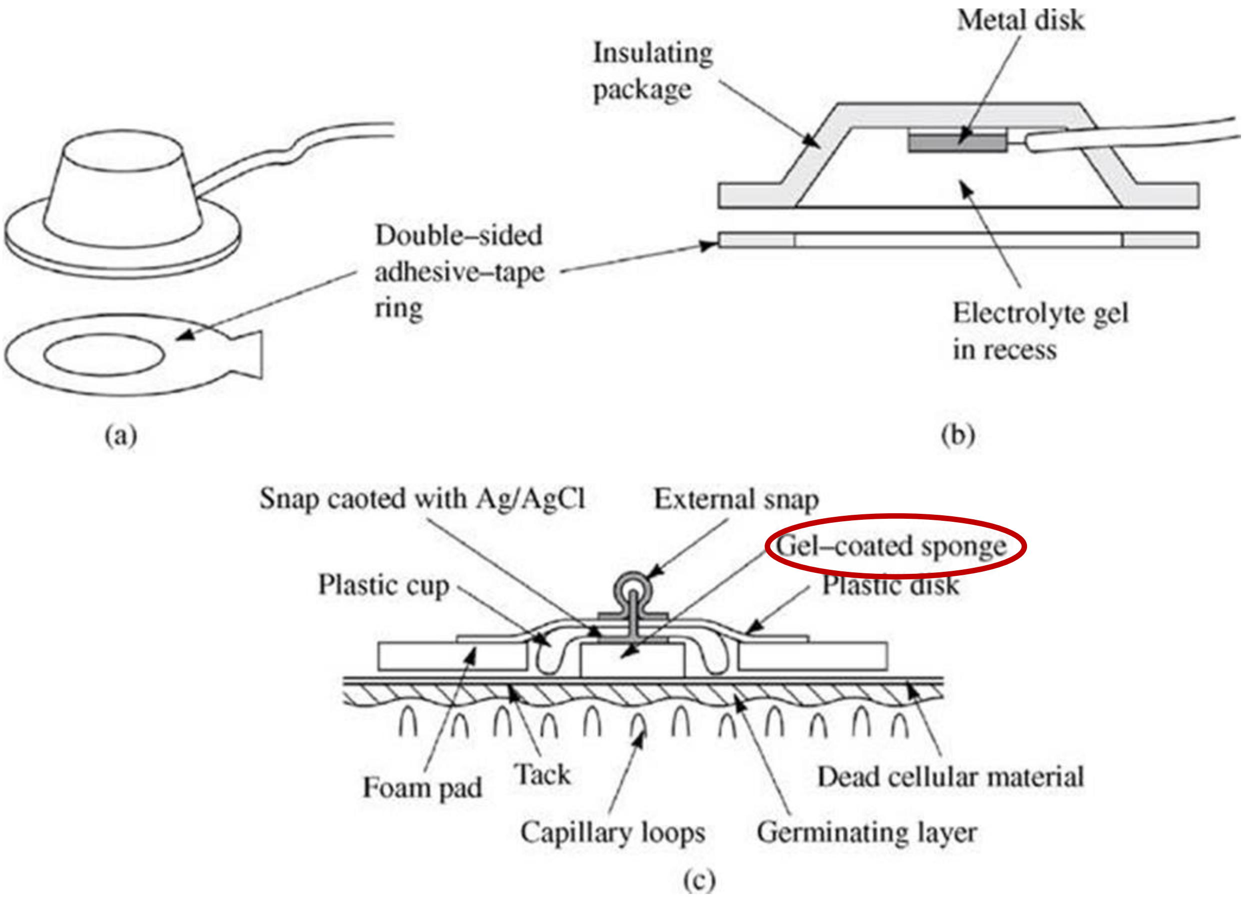
Metallic suction electrode

- A metallic suction electrode is often used as a precordial (chest) electrode on clinical electrocardiographs.
- It requires no straps or adhesives for holding it in place.
- Electrolyte gel is placed on the contacting surface of electrode.
- This electrode can be used only for short periods of time.



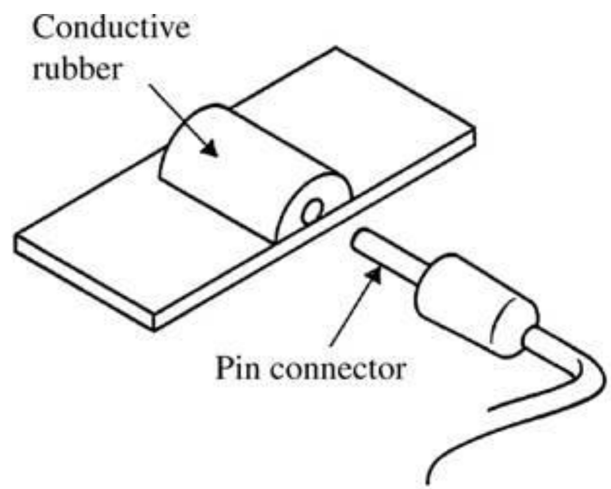
Floating Electrodes

- Mechanical technique to reduce noise. Isolates the electrode-electrolyte interface from motion artifacts.
- Metal disk (actual electrode) is recessed.
- Floating in the electrolyte gel.
- Not directly contact with the skin.
- Reduces motion artifacts.



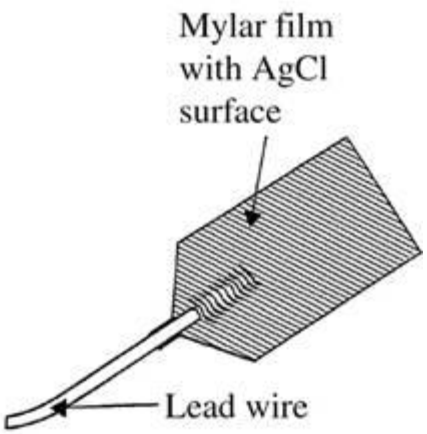
Flexible body-surface electrodes (a) Carbon-filled silicone rubber electrode

- Solid electrodes cannot conform to body-surface topology resulting additional motion artifacts.
- Carbon particles filled in the silicone rubber compound (conductive) in the form of a thin strip or disk is used as the active element of an electrode.
- A pin connector is pushed into the lead connector hole and electrode is usd like a metal plate electrode.
- Applications – monitoring premature infants (2500gr) who are not suitable for using standard electrodes.

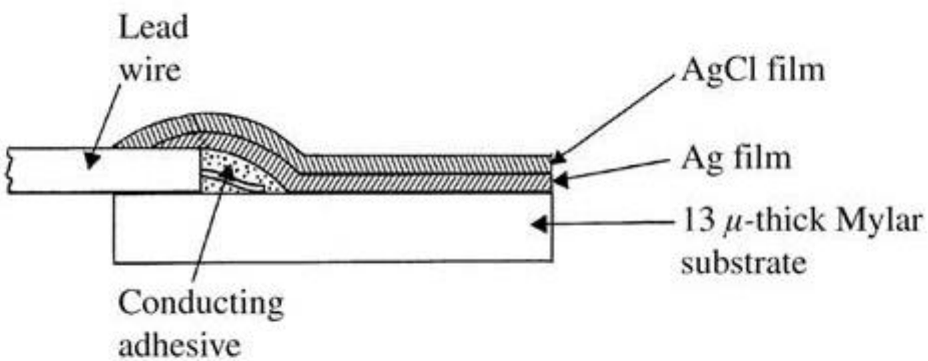


Flexible thin-film neonatal electrode (after Neuman, 1973). (c) Cross-sectional view of the thin-film electrode in (b)

- The basic electrode consists of 13 µm-thick Mylar film with Ag/AgCl film deposition.
- Lead wire is sticked to this film with an adhesive.
- Have the advantage of being flexible and conforming to the shape of newborn's chest.
- No need to be removed as they are X-ray transparent
- Monitoring new-born infants
- Drawback – High electric impeda



(b)

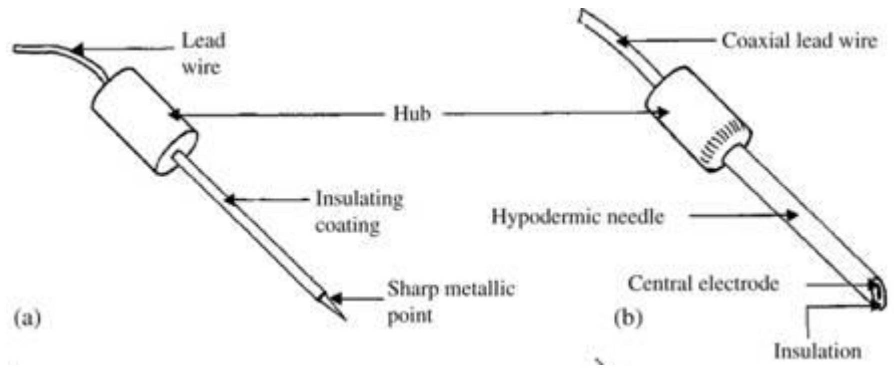


(c)

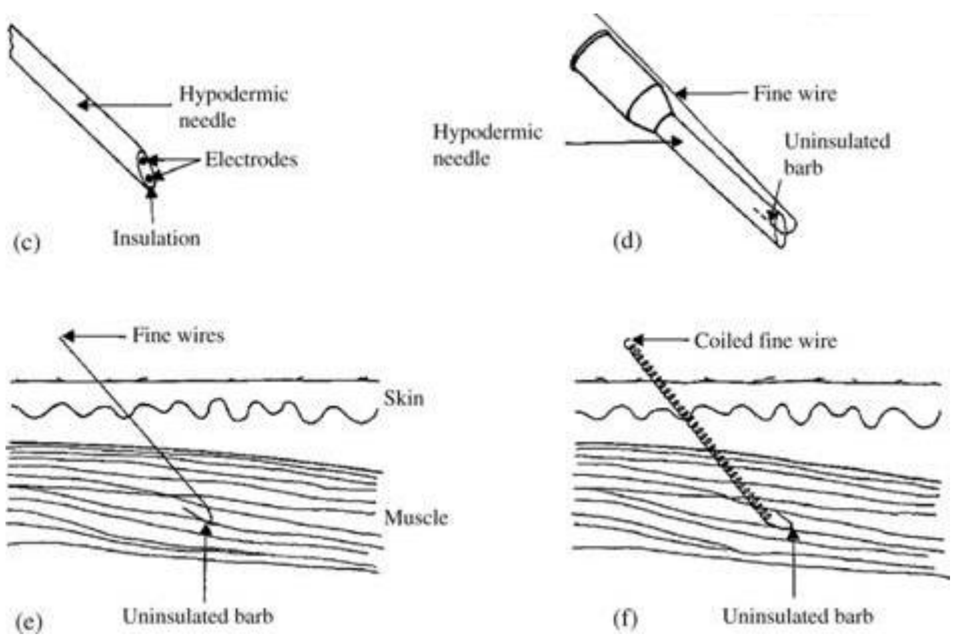
Internal Electrodes – detect biopotential within body

- *Percutaneous electrodes* - electrode itself or the lead wire crosses the skin,
- Entirely *internal electrodes* - connection is to an implanted electronic circuit such as a radiotelemetry transmitter.
- No limitation due to electrolyte-skin interface
- Electrode behaves in the way dictated entirely by the electrode–electrolyte interface.
- No electrolyte gel is required to maintain this interface, because extracellular fluid is present.

(a) Insulated needle electrode, (b) Coaxial needle electrode

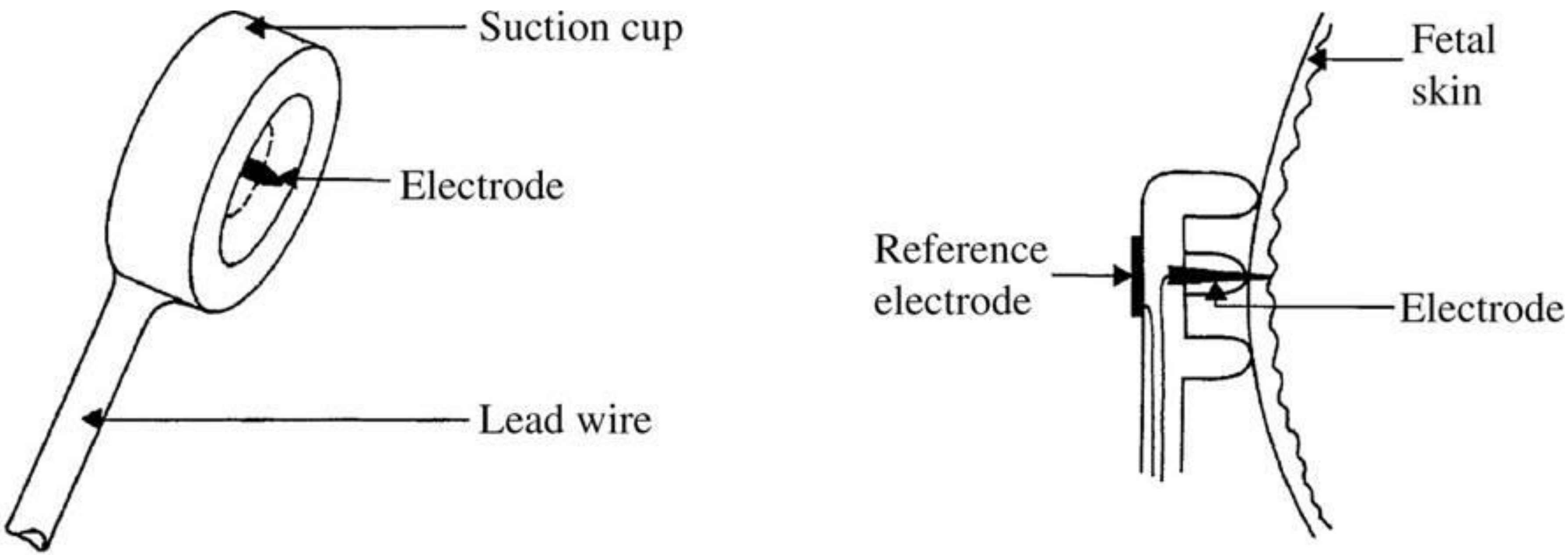


(c) Multiple electrodes in a single needle - Bipolar coaxial electrode,

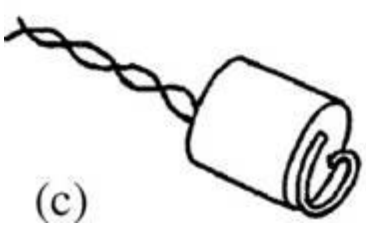


Chronic recordings using percutaneous wire electrodes - (d) Fine-wire electrode connected to hypodermic needle, before being inserted, (e) Cross-sectional view of skin and muscle, showing fine-wire electrode in place, (f) Cross-sectional view of skin and muscle, showing coiled fine-wire electrode in place.

Electrodes for detecting fetal electrocardiogram during labor, by means of intracutaneous needles (a) Suction electrode, (b) Cross-sectional view of suction electrode in place, showing penetration of probe through epidermis,

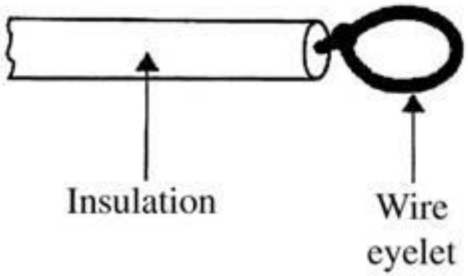


(a) (b)



(c)

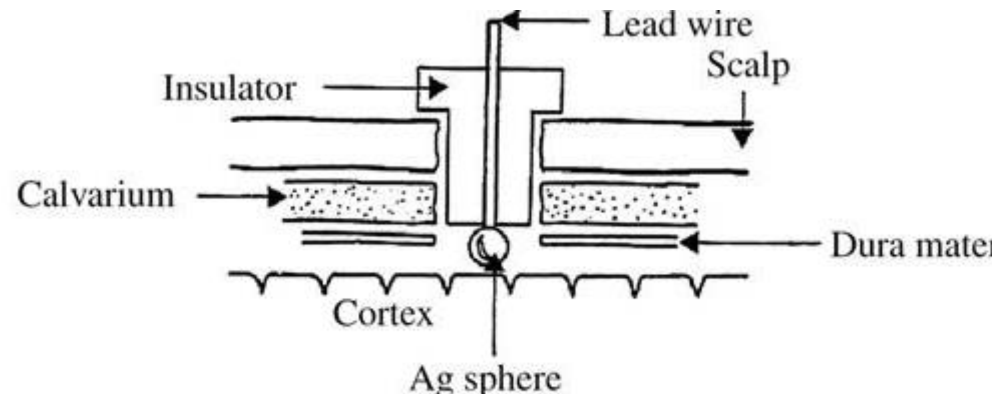
(c) Helical electrode, that is attached to fetal skin by corkscrew-type action.



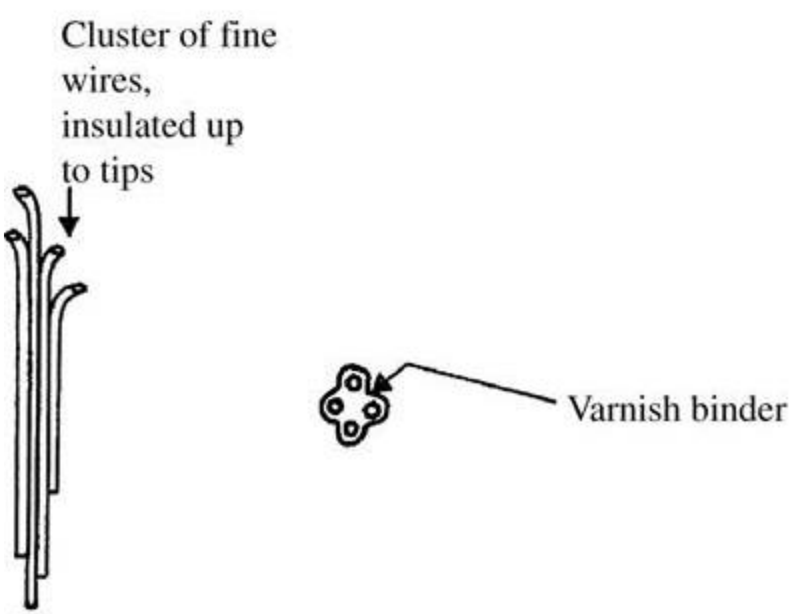
(a)

Insulated multistranded stainless steel or platinum wire wire-loop electrode for implantable wireless transmission electrodes for detecting biopotentials

(b) platinum-sphere cortical-surface potential electrode



(b)



(c)

(c) Multielement depth electrode for deep cortical potential measurement from multiple points.

Electrode Arrays

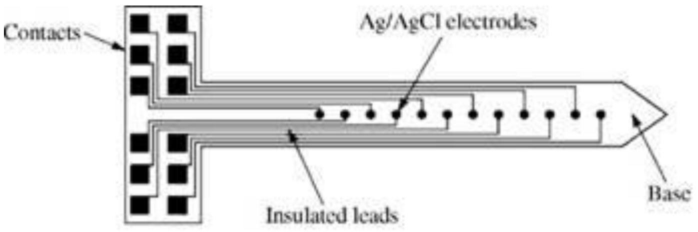
- Implantable electrode arrays can be fabricated one at a time using clusters of fine insulated wires, this technique is both time-consuming and expensive.
- When such clusters are made individually, each one will be somewhat different from the other.
- To minimize these problems is to utilize microfabrication technology to fabricate identical 2- and 3-D electrode arrays.

Examples of microfabricated electrode arrays,

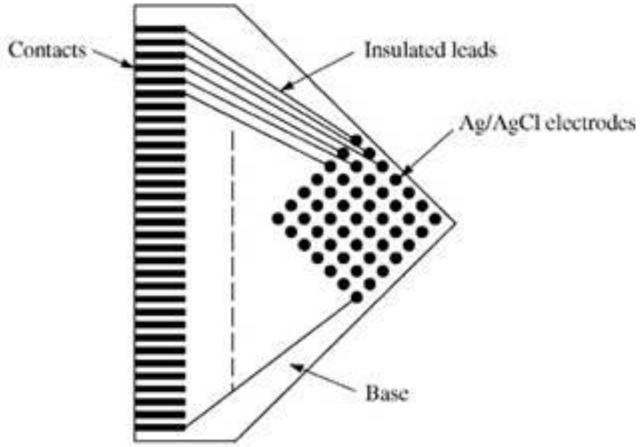
(a) One-dimensional plunge electrode array (after Mastrototaro *et al.*, 1992), -- measuring transmural potential distributions in beating myocardium

(b) Two-dimensional array – extracellular recording of neural signals in animal studies

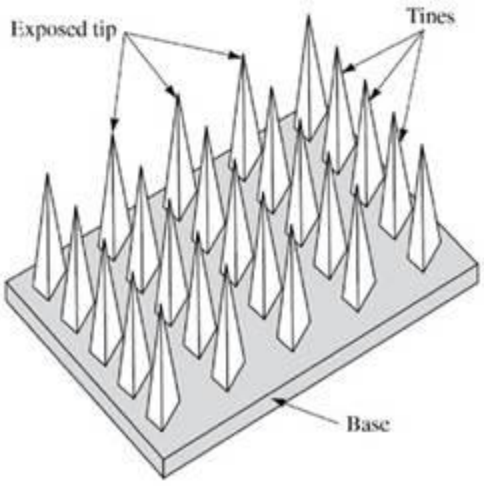
(c) Three-dimensional array (after Campbell *et al.*, 1991). – cardiac ECG mapping



(a)



(b)



(c)

Microelectrodes

Why

Measure potential difference across cell membrane

Requirements

- Small enough to be placed into cell
- Strong enough to penetrate cell membrane
- Typical tip diameter: 0.05 – 10 microns

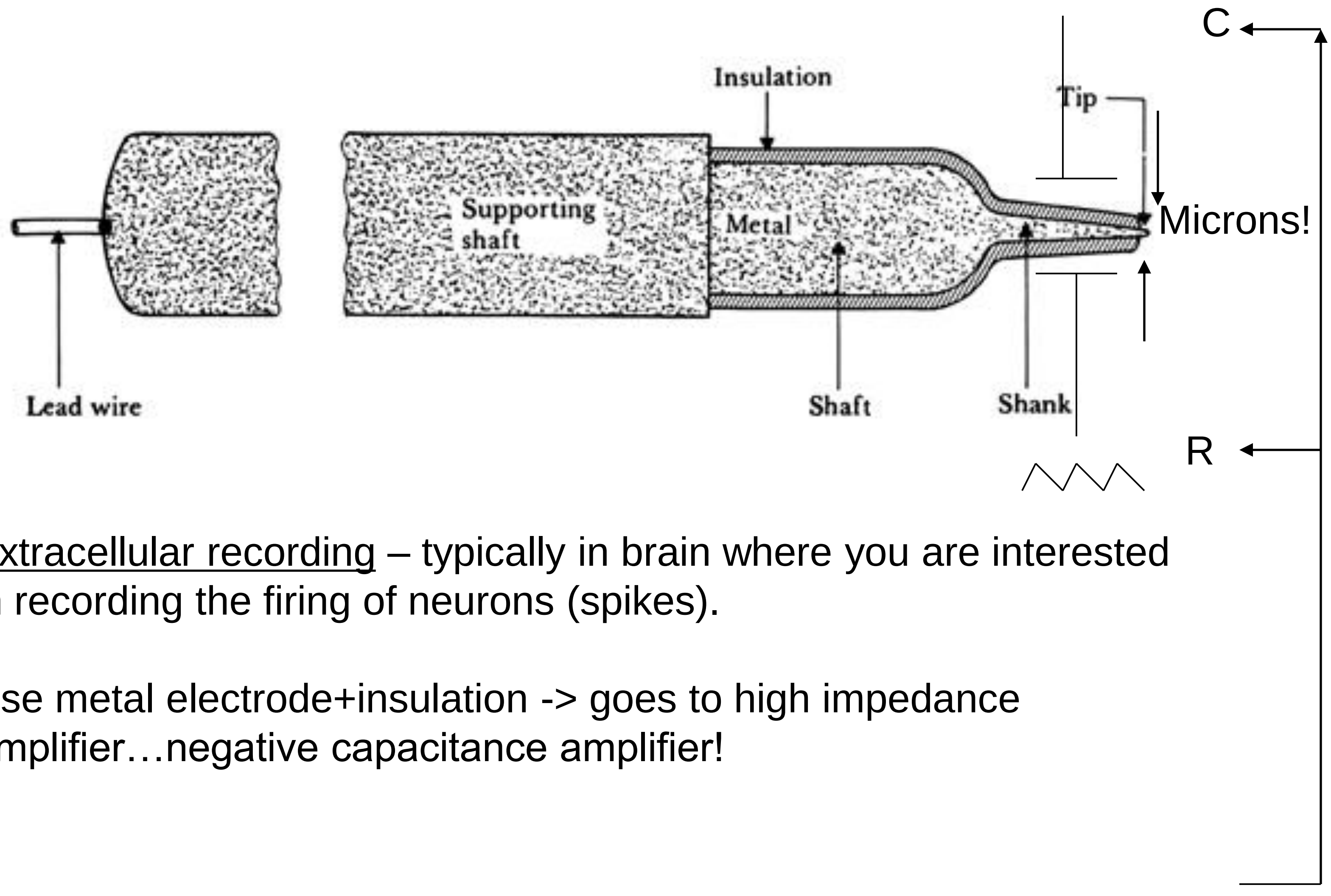
Intracellular

Extracellular

Types

- Solid metal -> Tungsten microelectrodes
- Supported metal (metal contained within/outside glass needle)
- Glass micropipette -> with Ag-AgCl electrode metal

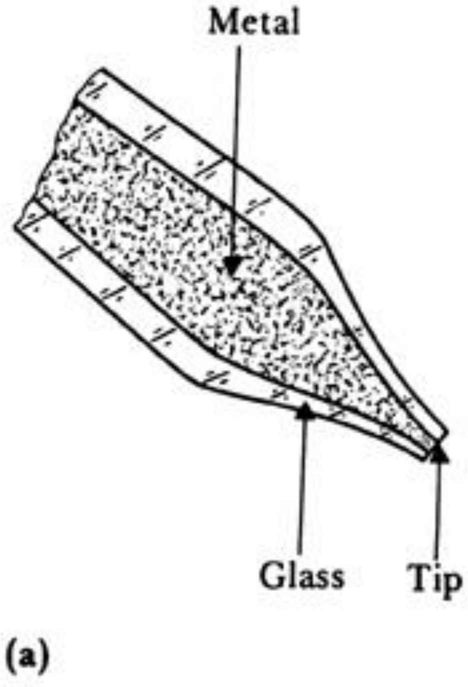
Metal Microelectrodes

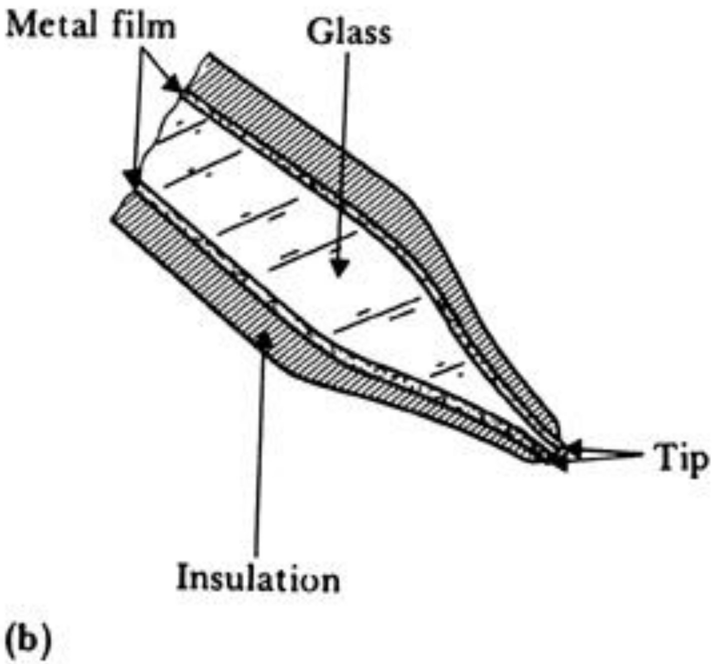


<u>Extracellular recording</u> – typically in brain where you are interested in recording the firing of neurons (spikes).

Use metal electrode+insulation -> goes to high impedance amplifier…negative capacitance amplifier!

Metal Supported Microelectrodes





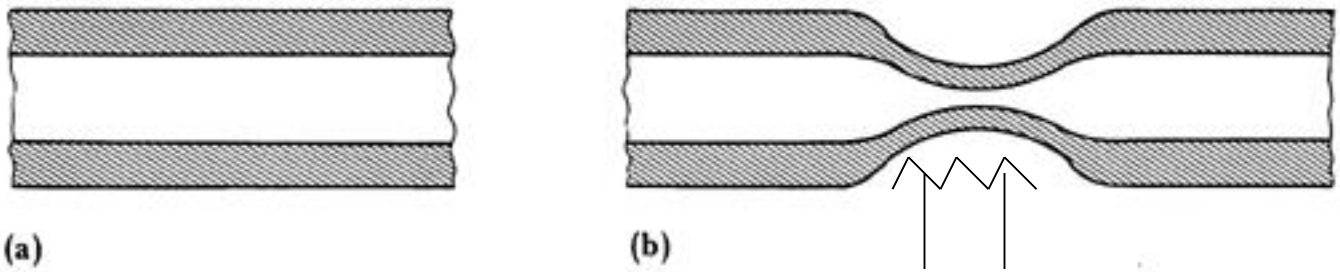
(a) Metal inside glass

(b) Glass inside metal

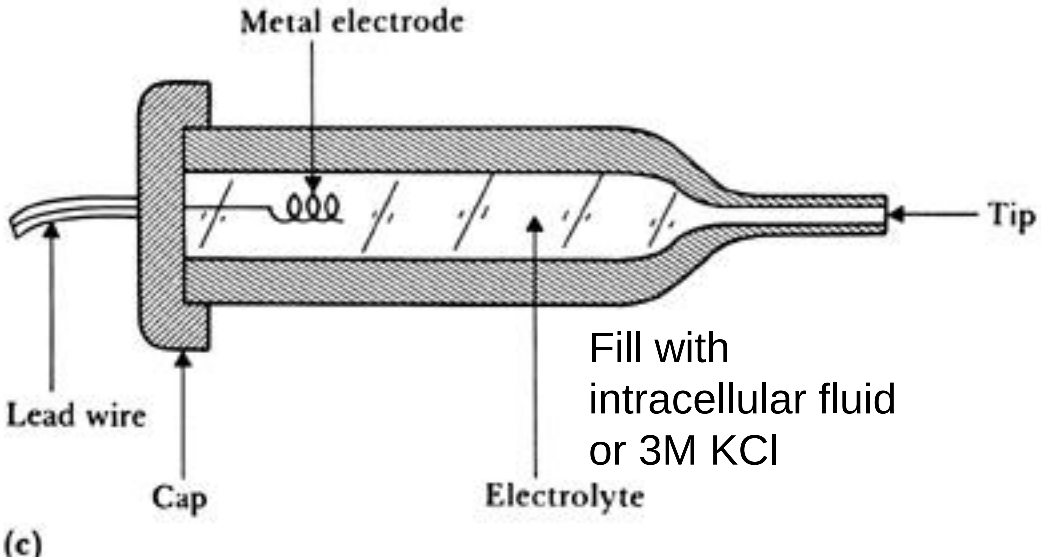
Glass Micropipette

Ag-AgCl wire+3M KCl has very low junction potential and hence very accurate for dc measurements (e.g. action potential)

heat



pull

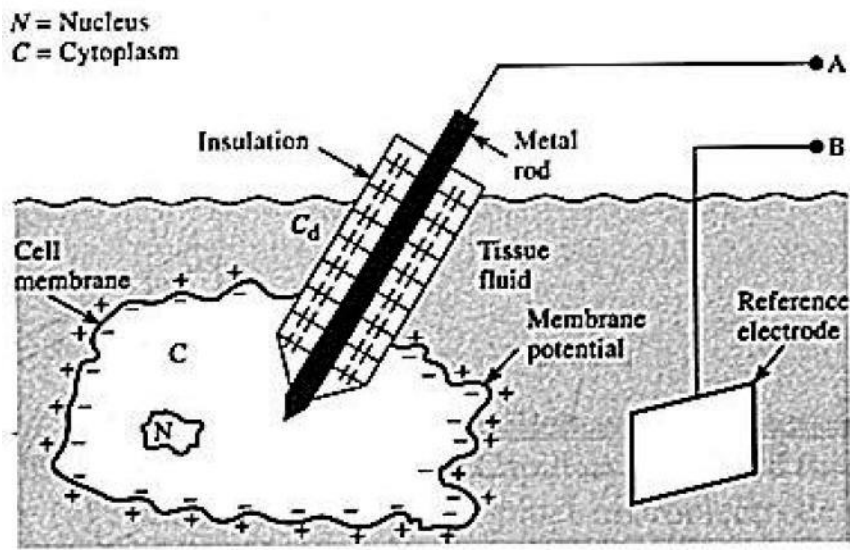


A glass micropipet electrode filled with an electrolytic solution
(a) Section of fine-bore glass capillary.
(b) Capillary narrowed through heating and stretching.
(c) Final structure of glass-pipet microelectrode.

<u>Intracellular recording</u> – typically for recording from cells, such as cardiac myocyte
<u>Need high impedance amplifier…negative capacitance amplifier</u>!

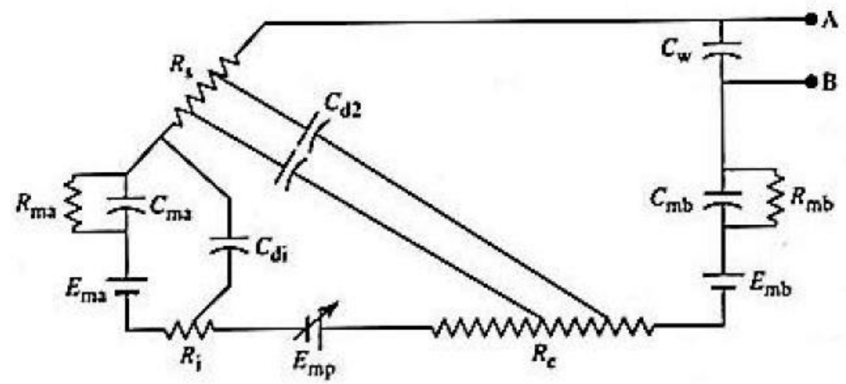
Electrical Properties of Microelectrodes

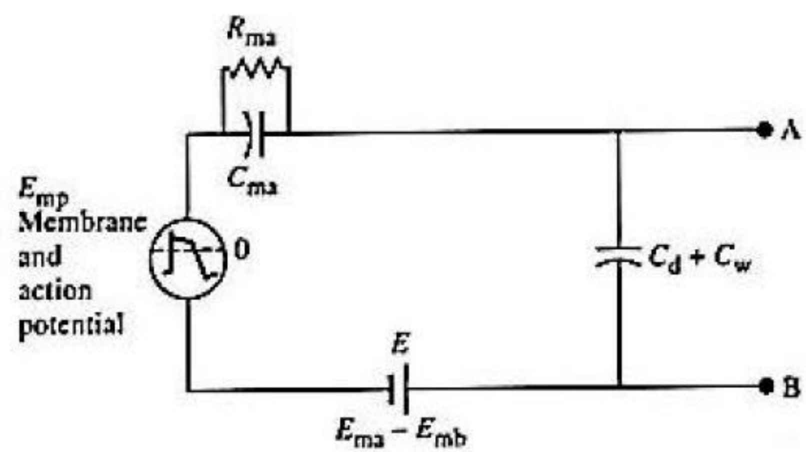
Metal Microelectrode



Metal microelectrode with tip placed within cell

Use metal electrode+insulation -> goes to high impedance amplifier…negative capacitance amplifier!

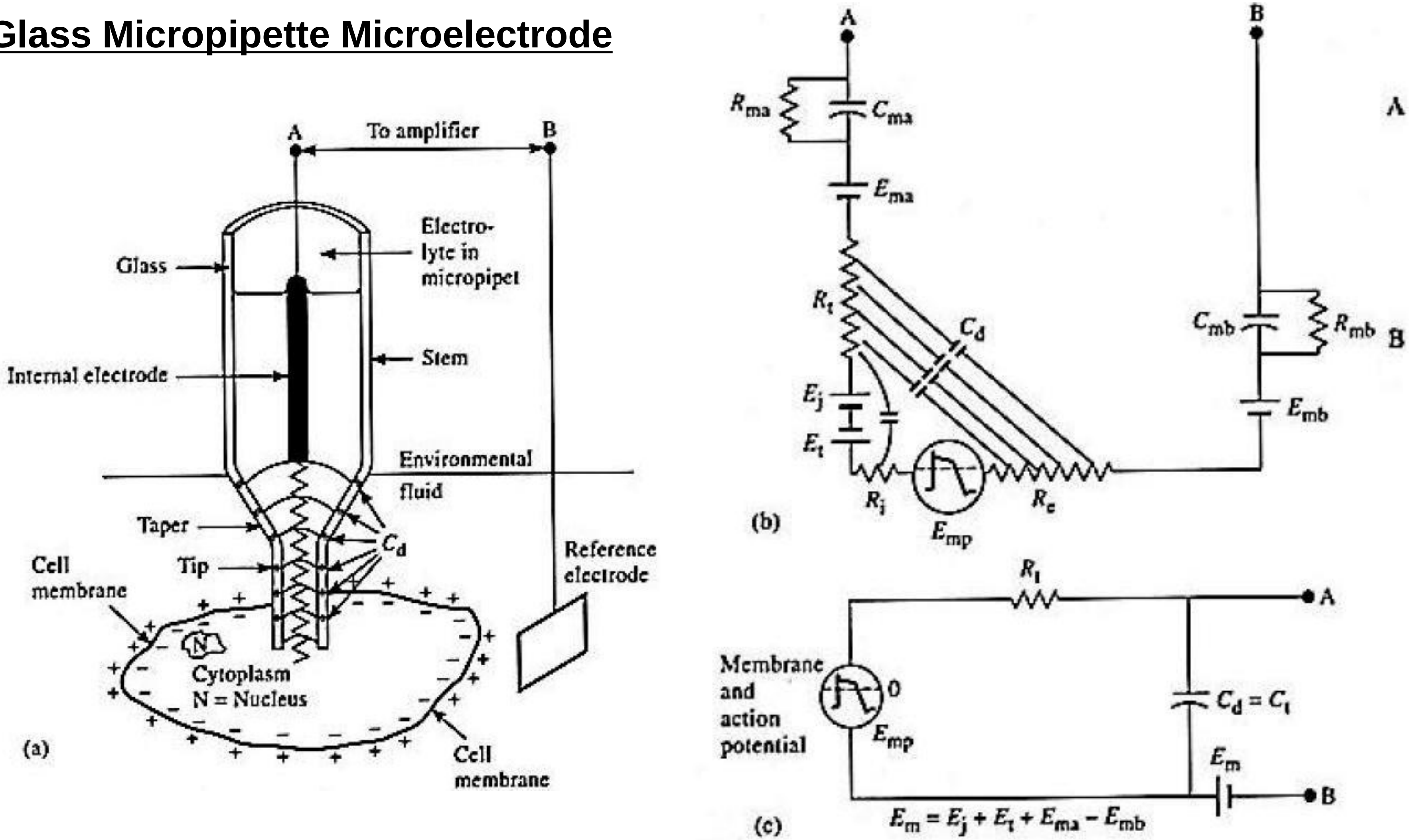




Equivalent circuits

Electrical Properties of Glass Intracellular Microelectrodes

Glass Micropipette Microelectrode



Stimulating Electrodes

Features

– Cannot be modeled as a series resistance and capacitance (there is no single useful model)
– The body/electrode has a highly nonlinear response to stimulation
– Large currents can cause
  – Cavitation
  – Cell damage
  – Heating

Types of stimulating electrodes

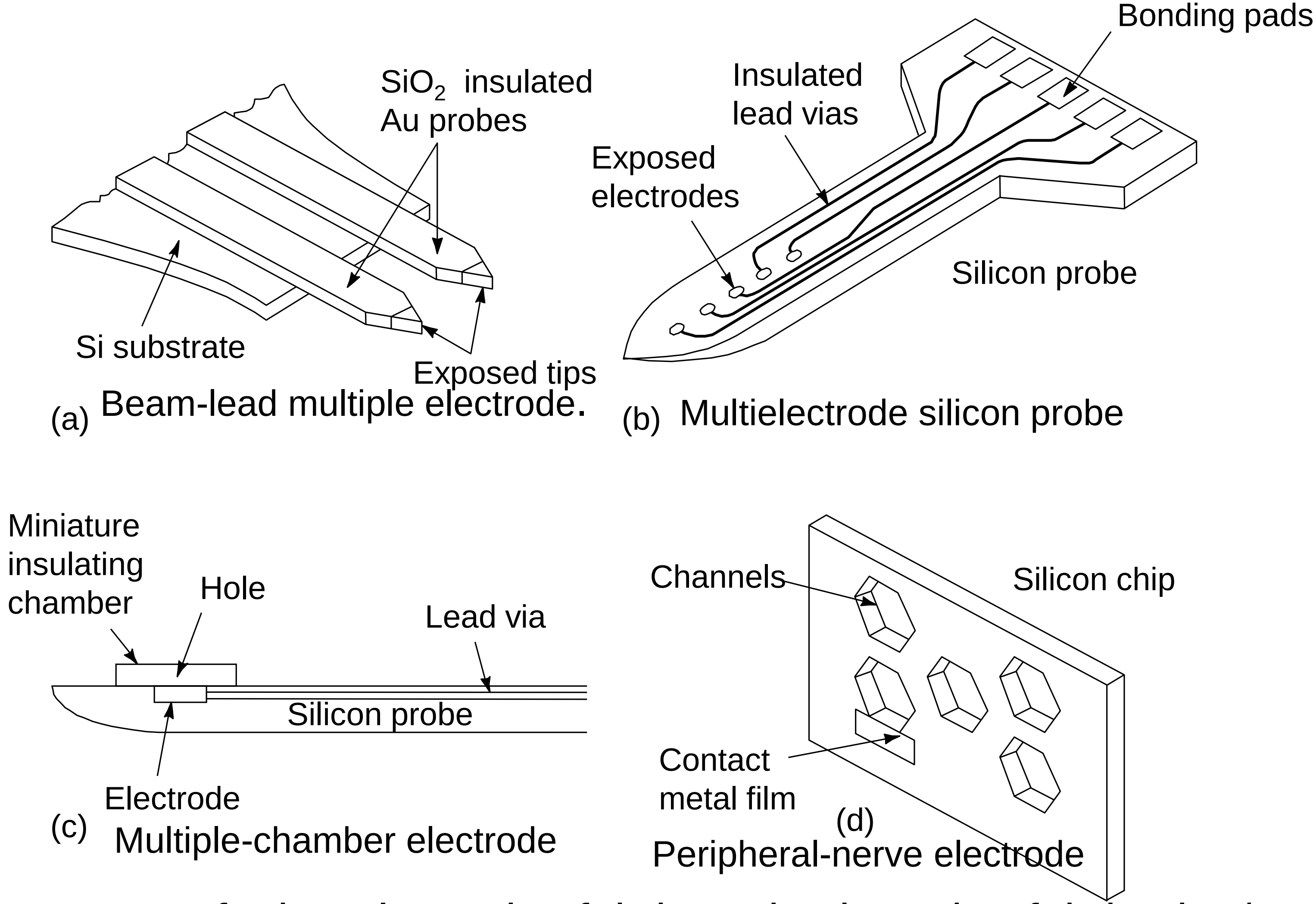
1. Pacing
2. Ablation
3. Defibrillation

Platinum electrodes:
Applications: neural stimulation

Modern day Pt-Ir and other exotic metal combinations to reduce polarization, improve conductance and long life/biocompatibility

Steel electrodes for pacemakers and defibrillators

Microelectronic technology for Microelectrodes



Different types of microelectrodes fabricated using microfabrication/MEMS technology

Michigan Probes for Neural Recordings

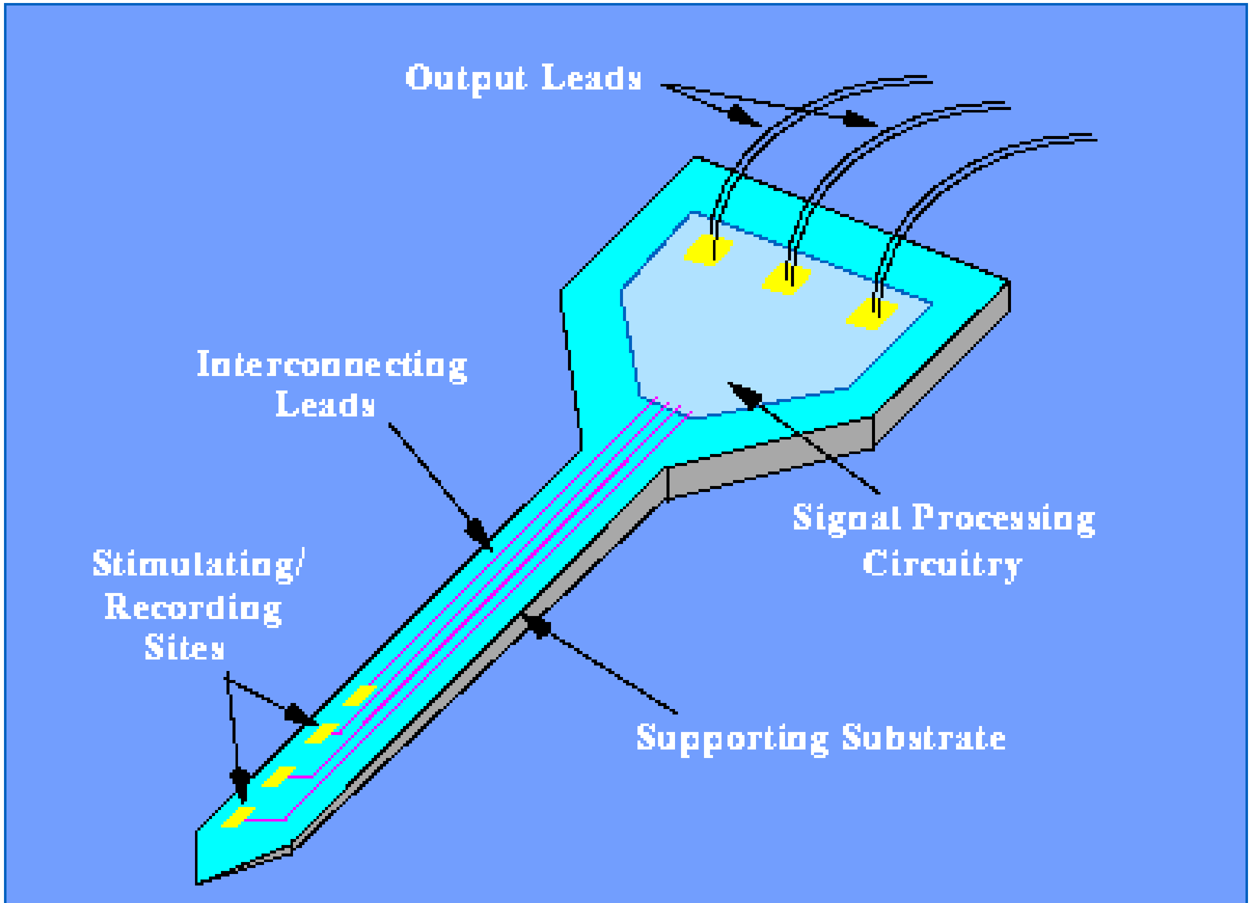


Figure 1: Basic structure of a micromachined multielectrode probe for recording and/or stimulation of the central nervous system.

Neural Recording Microelectrodes

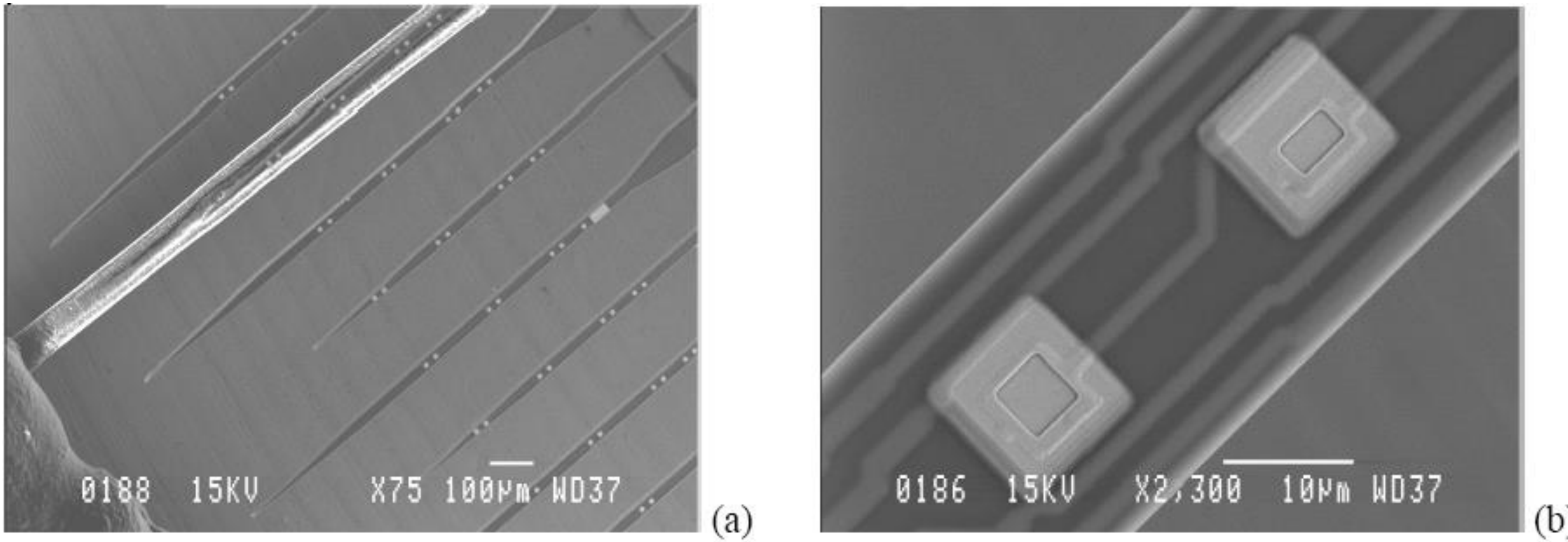


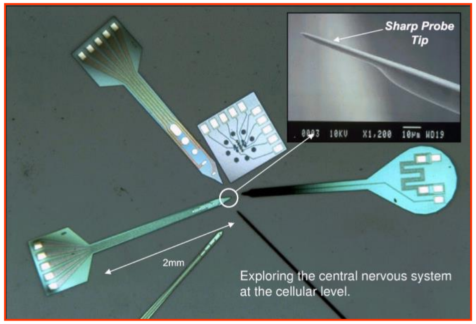
Figure 3. *(a) 8 shafts, each with 8 electrodes, compared to a human hair. (b) Close up of 2 electrodes.*

Reference :
http://www.acreo.se/acreo-rd/IMAGES/PUBLICATIONS/PROCEEDINGS/ABSTRACT-KINDLUNDH.PDF

In vivo neural microsystems: 3 examples

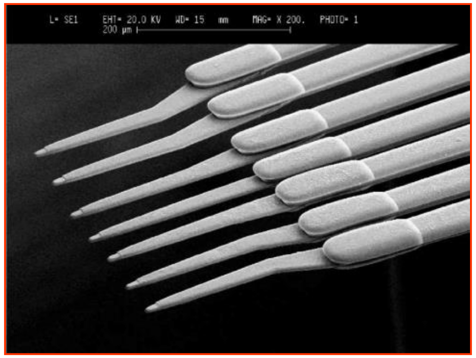
University of Michigan

Smart comb-shape microelectrode arrays for brain stimulation and recording



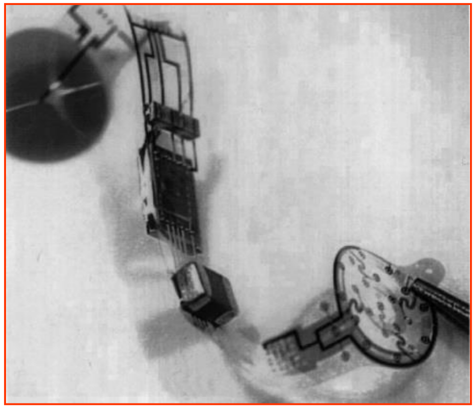
University of Illinois at Urbana-Champaign

High-density comb-shape metal microelectrode arrays for recording

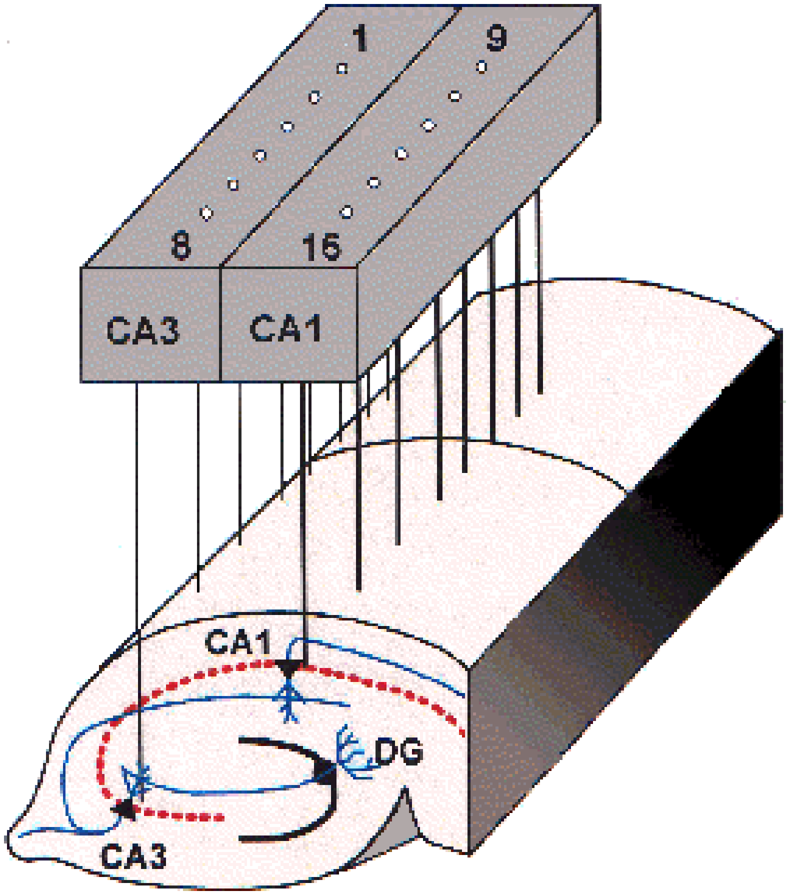


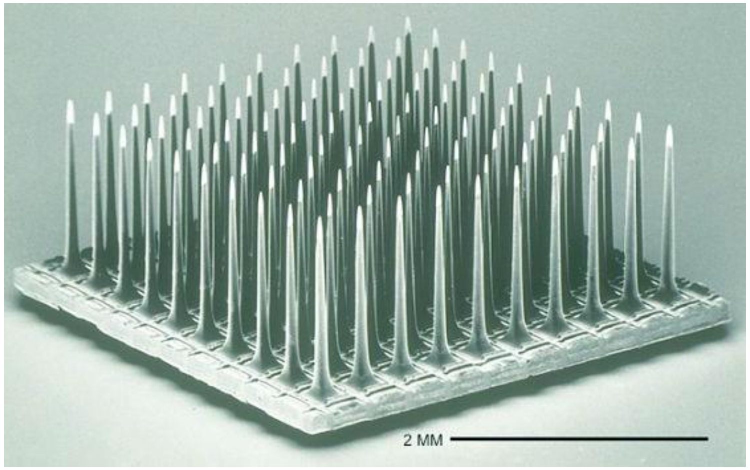
Fraunhofer Institute of Biomedical (FIBE) Engineering

Retina implant



Multi-electrode Neural Recording





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